

Full length article

Optimizing coal blending for metallurgical coke production using a reinforcement learning-enhanced evolutionary algorithm

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HIGHLIGHTS

- Proposed a multi-objective coal blending optimization model to minimize cost and maximize coke quality parameters.
- Introduced QNSGAIII, a Q-learning enhanced NSGA-III algorithm, outperforming five benchmark algorithms.
- Industrial case study demonstrates cost savings of 10–20 RMB/ton with optimized coal blend solutions.

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ABSTRACT

Achieving low-cost, high-quality coke production remains an ongoing challenge in the coking industry, where the coal blending optimization problem is recognized as NP-hard. Earlier approaches typically adopt single-objective optimization strategies, focusing on cost or selected quality indices but neglecting the need to balance both simultaneously. To address this challenge, we present a multi-objective coal blending optimization model (MOCB) that minimizes blending cost and maximizes critical coke quality parameters (ash, sulfur, M_{40} , M_{10} , CRI, CSR). We further propose a Q-learning enhanced Nondominated Sorting Genetic Algorithm-III (QNSGAIII) to effectively explore the extensive solution space—on the order of 100^{20} when blending 20 types of coal—while maintaining the required process constraints. Benchmark tests on DTLZ and WFG functions show that QNSGAIII surpasses five classical champion algorithms (MOEA/D, PESA-II, RVEA, NSGA-II, NSGA-III) in terms of convergence and diversity (IGD, HV, etc.). An industrial case study involving three distinct metallurgical coke products demonstrates how the QNSGAIII-derived Pareto-optimal solutions simultaneously achieve cost savings and meet quality specifications, revealing robust and diverse coal blend structures. Optimization of coal blending reduces the use of premium coking and fat coals by 3 %–5 % and lowers blending costs by 10–20 RMB per ton. These results demonstrate the practical value of incorporating reinforcement learning into evolutionary algorithms for multi-objective coal blending, providing a route to more efficient and intelligent cokemaking processes.

1. Introduction

The continuous growth of the global steel industry has led to an increasing demand for coking coal [1,2]. Although the market size is influenced by various unstable factors, the significance of coking coal in steel production ensures its robust market outlook [3,4]. Currently, coking coal accounts for approximately 13 % of the global coal resources, with an estimated total of 11,400 billion tons [5,6]. Notably, 48.5 %

of this is distributed across Europe, 29 % in Asia, and 19 % in North America [7]. China, the world's largest producer and consumer of coking coal, holds 26 % of the global reserves, with its coking technology and equipment exported to over 10 countries [8,9]. While the remaining reserves in major production areas are diminishing rapidly [10,11], the need for more scientifically-driven development and utilization of domestic and global coking coal resources is becoming a pressing challenge for coking enterprises [12,13].

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Nomenclature

Blending and coke quality indicators

Ad	Ash content (dry basis, %)
St, d	Total sulfur content (dry basis, %)
V_{daf}	Volatile matter (dry ash-free basis, %)
G	Maximum dilatation or caking index of coal
Y	Maximum plastic layer thickness (mm)
M_{40}	Coke crushing strength (%)
M_{10}	Coke abrasion strength (%)
CRI	Coke Reactivity Index (%)
CSR	Coke Strength after Reaction (%)

Optimization algorithms and performance metrics

NSGA-II	Nondominated Sorting Genetic Algorithm II
NSGA-III	Nondominated Sorting Genetic Algorithm III
QNSGAIII	Q-learning enhanced NSGA-III
MOEA/D	Multi-Objective Evolutionary Algorithm based on Decomposition
PESA-II	Pareto Envelope-based Selection Algorithm II
RVEA	Reference Vector Guided Evolutionary Algorithm
RL	Reinforcement Learning
BPNN	Backpropagation Neural Network
HV	Hypervolume indicator
IGD	Inverted Generational Distance

Coking coal is primarily used in coke production, which in turn is crucial for the processes of ironmaking and steelmaking [14,15]. Metallurgical coke, as an essential fuel and raw material in steel manufacturing [16–18], plays a significant role in controlling energy consumption within both the steel and coking industries through optimization of its production process [19,20]. Given the increasing size of blast furnaces and the introduction of oxygen-enriched pulverized coal injection [21,22], coke's role as a permeable support structure has become even more critical, placing higher demands on its quality [23]. To ensure the quality of pig iron, a blend of various coals is typically used in the coking process, which maximizes the advantages of each coal type [24]. This blending process ensures that coke quality parameters are met, while also optimizing the use of high-quality coking coal and extending the available coal resources [25,26]. Consequently, coke quality prediction and coal blending optimization have become key focus areas for coking enterprises.

Coke quality prediction and coal blending optimization are fundamentally interdependent processes that together form the technical core for improving efficiency and product consistency in the coking industry [27–29]. The ability to accurately predict key coke quality indices—such as mechanical strength (M_{40} , M_{10}), reactivity index (CRI), and coke strength after reaction (CSR)—is essential for guiding optimal coal blending [30–32]. These indices are the outcome of complex, nonlinear interactions among multiple coal quality parameters, including volatile matter, caking index, and plasticity, which are difficult to capture with conventional approaches based on weighted averages or empirical formulas [33,34]. As blast furnace operations evolve toward larger scales and higher technical requirements, traditional prediction methods have become insufficient to meet the increasingly stringent quality standards. In response, data-driven machine learning models such as artificial neural networks (ANNs) and support vector machines (SVMs) have demonstrated superior capability in modeling these nonlinear relationships, learning intricate dependencies between coal blend characteristics and coke properties [35,36]. The adoption of such advanced predictive tools provides a robust scientific basis for coal blending optimization, as accurate coke quality predictions can serve directly as objective functions and constraints within optimization frameworks. This integration allows coking enterprises to systematically adjust coal blends, lower costs, and reduce reliance on premium coking coals, all while maintaining required product standards [37]. Therefore, advancing high-accuracy coke quality prediction technologies is central to realizing intelligent and precise coal blending optimization [38], which underpins stable production and economic performance in the coking process.

Coal blending optimization refers to the process of selecting the best mix of coals under given constraints to minimize costs and maximize coke quality, which is inherently challenging due to the diverse varieties, inconsistent quality, and varying inventory levels of single coals. These factors complicate the calculation of optimal blending ratios, as

the constraints involved are complex and multifaceted, including required ranges of parameters such as ash content, volatile matter, sulfur content, and mechanical strength, as well as external factors like coal inventory and supply availability [39,40]. Traditional manual blending methods, reliant on empirical experience, often fail to accurately predict the quality indicators of blended coal, leading to significant fluctuations in coke quality and higher production costs. Consequently, coal blending optimization is a multi-objective problem that requires advanced computational approaches to balance competing constraints and achieve both economic and quality objectives.

Numerous data-driven methods have been proposed to address the complexities of coal blending optimization. Typically, these approaches formulate constraint-based optimization problems and employ advanced algorithms, such as genetic algorithms (GA) or particle swarm optimization (PSO) [41], to derive optimal blending schemes. These techniques have significantly enhanced the efficiency and intelligence of coal blending solutions [42,43]. However, a notable limitation of existing methods is their predominant focus on single-objective optimization, often prioritizing cost minimization. In practice, coal blending is a multi-faceted problem that requires balancing cost efficiency with critical quality parameters, such as ash and sulfur content, to ensure optimal coke performance. To address this, multi-objective optimization models have been developed, integrating targets such as cost reduction, sulfur content minimization, and calorific value maximization. Despite their advantages, these models face practical challenges, particularly in determining appropriate weightings for each objective, as the relative importance of parameters like cost, quality, and environmental impact is often difficult to quantify and may vary depending on specific operational priorities. Consequently, while significant progress has been made, further research is needed to refine multi-objective optimization frameworks for coal blending, ensuring they are both theoretically robust and practically applicable.

Recent advances in multi-objective coal blending optimization have demonstrated promising approaches to balance cost and quality constraints. Hybrid methodologies integrating artificial neural networks with evolutionary algorithms have shown particular efficacy, with [44] achieving adaptive blend formulation through BP neural network-guided genetic algorithms. Similarly, [45] developed a simulated annealing-based framework that simultaneously minimizes sulfur content while maximizing calorific value. These approaches address the inherent nonlinearities in coal-to-coke transformations more effectively than traditional weighted-average models [46]. However, most studies focus predominantly on reactivity indices (CRI/CSR), often neglecting comprehensive mechanical strength optimization (M_{40}/M_{10}) despite their critical role in blast furnace performance [37,47].

Significant challenges persist in operational deployment of optimization frameworks. Industrial implementations frequently encounter difficulties in dynamically adjusting to raw material variability and process constraints [48]. Fixed parameter settings in evolutionary

algorithms limit adaptability across diverse production scenarios, while manual weight assignment in multi-objective models introduces subjectivity [49]. Furthermore, few studies validate optimization outcomes across full-scale coke oven operations, creating gaps between theoretical models and practical application [50]. These limitations underscore the need for intelligent optimization mechanisms capable of self-adjusting to dynamic production environments while holistically addressing both economic and metallurgical requirements.

In recent years, heuristic and meta-heuristic algorithms have gained significant traction in optimizing complex industrial systems, such as thermal power plants, due to their ability to handle intricate constraints and multi-objective requirements. Among these, evolutionary algorithms, a prominent subset of meta-heuristics, stand out for their theoretical simplicity and computational efficiency. Notable examples include the Multi-Objective Evolutionary Algorithm Based on Decomposition (MOEA/D) [51], the Pareto Envelope-based Selection Algorithm-II (PESA-II) [52], the Reference Vector Guided Evolutionary Algorithm (RVEA) [53], and the Nondominated Sorting Genetic Algorithm series (NSGA-II and NSGA-III) [54–56]. Specifically, NSGA-III has emerged as a powerful tool for multi-objective optimization, leveraging reference points in its selection mechanism to achieve a more uniform Pareto front distribution. This enhancement significantly improves both the convergence and diversity of solutions, making NSGA-III particularly suitable for complex optimization tasks. However, despite its proven effectiveness in various domains, its application to coal blending optimization in metallurgical coke production remains underexplored. A critical challenge in applying NSGA-III to coal blending lies in the dynamic tuning of key parameters, such as crossover probability (P_c) and mutation probability (P_m), which directly influence the algorithm's performance. Early in the optimization process, higher values of these parameters can enhance population diversity and accelerate convergence, while later stages require careful reduction to preserve high-quality solutions and avoid stagnation.

To overcome these limitations and further enhance NSGA-III's search capabilities, this study proposes the integration of reinforcement learning (RL), specifically Q-Learning, for adaptive parameter tuning. Reinforcement learning, particularly double Q-Learning, has demonstrated remarkable success in optimizing algorithm performance and adaptability in dynamic environments [57,58]. By leveraging historical and real-time data, Q-Learning generates optimal decision-making strategies [59,60], making it highly effective for fine-tuning key parameters in optimization algorithms. Recent studies have highlighted the potential of Q-Learning in improving convergence accuracy and real-time adaptability in complex scheduling tasks, such as workshop production systems [61,62]. Building on these advancements, this paper introduces a novel hybrid approach, termed Q-learning enhanced NSGA-III(QNSGAIII), tailored specifically for coal blending optimization in metallurgical coke production. This method aims to address the dual challenges of balancing multiple objectives—such as cost minimization, sulfur content reduction, and calorific value maximization—while dynamically adjusting algorithm parameters to ensure robust and efficient performance. By integrating Q-Learning with NSGA-III, this study seeks to provide a more intelligent and adaptive solution to the complex coal blending problem, ultimately enhancing both the quality and cost-effectiveness of metallurgical coke production.

The primary innovations of this work are threefold. First, this study develops a reinforcement learning-enhanced evolutionary algorithm, namely a QNSGAIII, for multi-objective coal blending optimization. The proposed QNSGAIII enables adaptive parameter adjustment and operator selection, significantly improving the convergence, diversity, and robustness of the search process in complex, high-dimensional optimization tasks. Its effectiveness is rigorously validated using standard multi-objective benchmark functions, including the DTLZ and WFG series, where QNSGAIII is comprehensively compared with five state-of-the-art algorithms (MOEA/D [51], PESA-II [52], RVEA [53], NSGA-II [54], NSGA-III [55]) using IGD and HV indicators to assess convergence

and diversity performance. Second, this work establishes an integrated cost–quality collaborative optimization framework, which couples a high-accuracy BPNN model for coke quality prediction with multi-objective coal blending optimization. This integration allows for precise mapping of nonlinear relationships between blend characteristics and critical coke quality indices, enabling coordinated cost minimization and quality control within practical constraints. Third, the proposed method is extensively validated on both laboratory and industrial coke oven datasets, covering up to 20 coal types and three grades of metallurgical coke. Comparative experiments with the five benchmark algorithms in these real-world scenarios further demonstrate the robustness, generalizability, and practical superiority of the proposed approach for coal blending optimization in modern cokemaking practice.

The remainder of this paper is organized as follows. Section 2 presents the formulation of the multi-objective coal blending optimization problem and details the integration of the ANN-based coke quality prediction model. Section 3 describes the proposed QNSGAIII algorithm and outlines the benchmark settings for algorithm comparison. In Section 4, we report the results of comparative tests on DTLZ and WFG benchmark functions using IGD and HV metrics, followed by application and validation in laboratory- and industrial-scale coke production scenarios. Section 5 discusses the practical implications, limitations, and future directions for intelligent coal blending optimization. Finally, Section 6 summarizes the main findings and contributions of this work.

2. Multi-objective coal-blending optimisation model

Coal blending is formulated as a constrained multi-objective problem. The decision vector comprises the mass fractions of the available coals, subject to inventory and throughput limits. Seven objectives are optimised concurrently: unit cost, ash, total sulphur, crushing strength, abrasion index, coke reactivity index and coke strength after reaction.

2.1. Objective functions

2.1.1. Objective 1: minimizing blending cost

The first objective is to minimize the blending cost, denoted as P_{blend} . The blending cost is determined by the weighted sum of the individual coal costs:

$$P_{\text{blend}} = \sum_{i=1}^n P_i x_i, \quad (1)$$

where P_i represents the cost of the i -th coal type, and x_i is the proportion of the i -th coal in the blend.

2.1.2. Objective 2: minimizing ash content of coke

The second objective is to minimize the ash content of the coke produced from the coal blend, denoted as A_{blend} . The ash content is calculated as the weighted average of the ash contents of the individual coals' cokes:

$$A_{\text{blend}} = \sum_{i=1}^n A_{d,i} x_i + b_A, \quad (2)$$

where $A_{d,i}$ represents the ash content of the coke produced from the i -th coal type, and b_A is a correction factor.

2.1.3. Objective 3: minimizing sulfur content of coke

The third objective is to minimize the sulfur content of the coke, denoted as S_{blend} . The sulfur content is calculated as the weighted average of the sulfur contents of the individual coals' cokes:

$$S_{\text{blend}} = \sum_{i=1}^n S_{t,d,i} x_i + b_s, \quad (3)$$

where $S_{t,d,i}$ represents the sulfur content of the coke produced from the i -th coal, and b_s is a correction factor.

2.1.4. Objective 4: maximizing coke crushing strength

The fourth objective is to maximize the M_{40} value, which represents the crushing strength of the coke. The M_{40} value is predicted using a Backpropagation neural network (BPNN) model. The input to the neural network includes the weighted average values of V_{daf} , G , and Y of the blended coal, as well as the weighted average M_{40} values of the individual coals produced from the blend. The output of the neural network is the predicted M_{40} value for the blended coke. The relationship is expressed as:

$$M_{40\text{blend}} = f_{M_{40}}(\overline{V_{daf}}, \overline{G}, \overline{Y}, \overline{M_{40}}), \quad (4)$$

where $\overline{V_{daf}}$, $\overline{G}$, $\overline{Y}$, and $\overline{M_{40}}$ represent the weighted average values of volatile matter, caking index, maximum plastic layer, and M_{40} from the individual coals, respectively. $f_{M_{40}}$ is the BPNN model that predicts the M_{40} value for the blended coke.

2.1.5. Objective 5: minimizing coke abrasion strength

The fifth objective is to minimize the M_{10} value, which represents the abrasion strength of the coke. The M_{10} value is predicted using a BP neural network model. The input to the neural network includes the weighted average values of V_{daf} , G , and Y of the blended coal, as well as the weighted average M_{10} values of the individual coals produced from the blend. The output of the neural network is the predicted M_{10} value for the blended coke. The relationship is expressed as:

$$M_{10\text{blend}} = f_{M_{10}}(\overline{V_{daf}}, \overline{G}, \overline{Y}, \overline{M_{10}}), \quad (5)$$

where $\overline{M_{10}}$ represents the weighted average values of volatile matter, caking index, maximum plastic layer, and M_{10} from the individual coals, respectively. $f_{M_{10}}$ is the BPNN model that predicts the M_{10} value for the blended coke.

2.1.6. Objective 6: minimizing coke reactivity index

The sixth objective is to minimize the CRI value, which represents the reactivity of the coke. The CRI value is predicted using a BPNN model. The input to the neural network includes the weighted average values of V_{daf} , G , and Y of the blended coal, as well as the weighted average CRI values of the individual coals produced from the blend. The output of the neural network is the predicted CRI value for the blended coke. The relationship is expressed as:

$$CRI_{\text{blend}} = f_{CRI}(\overline{V_{daf}}, \overline{G}, \overline{Y}, \overline{CRI}), \quad (6)$$

where $\overline{CRI}$ represents the weighted average values of volatile matter, caking index, maximum plastic layer, and CRI from the individual coals, respectively. f_{CRI} is the BPNN model that predicts the CRI value for the blended coke.

2.1.7. Objective 7: maximizing coke strength after reaction

The seventh objective is to maximize the CSR value, which represents the coke strength after reaction. The CSR value is predicted using a BP neural network model. The input to the neural network includes the weighted average values of V_{daf} , G , and Y of the blended coal, as well as the weighted average CSR values of the individual coals produced from the blend. The output of the neural network is the predicted CSR value for the blended coke. The relationship is expressed as:

$$CSR_{\text{blend}} = f_{CSR}(\overline{V_{daf}}, \overline{G}, \overline{Y}, \overline{CSR}), \quad (7)$$

where $\overline{CSR}$ represents the weighted average values of volatile matter, caking index, maximum plastic layer, and CSR from the individual coals, respectively. f_{CSR} is the BPNN model that predicts the CSR value for the blended coke.

2.2. Constraints

In the optimization of coal blending, several constraints must be considered, including the blending ratio of each coal type, the quality of the blended coal (and the produced coke), and process constraints. The constraints are expressed as follows:

$$\begin{aligned} \sum_{j=1}^n x_j &= 100\%, \quad x_j \geq 0 \\ \text{s.t.} \quad \overline{A_{\text{blend}}} &\in [A_{d\min}, A_{d\max}], \quad \overline{S_{\text{blend}}} \in [S_{(t,d)\min}, S_{(t,d)\max}], \\ \overline{M_{40\text{blend}}} &\in [M_{40\min}, M_{40\max}], \quad \overline{M_{10\text{blend}}} \in [M_{10\min}, M_{10\max}], \\ \overline{CRI_{\text{blend}}} &\in [CRI_{\min}, CRI_{\max}], \quad \overline{CSR_{\text{blend}}} \in [CSR_{\min}, CSR_{\max}], \\ x_j &\in [x_{j\min}, x_{j\max}], \end{aligned}$$

where the sum of the blending ratios of all coal types must equal 100 %, and the actual blending ratio of each coal type must be non-negative. The variables $\overline{A_{\text{blend}}}$, $\overline{S_{\text{blend}}}$, $\overline{M_{40\text{blend}}}$, $\overline{M_{10\text{blend}}}$, $\overline{CRI_{\text{blend}}}$, and $\overline{CSR_{\text{blend}}}$ represent the quality indicators of the coke produced from the blended coal, and these values must lie within the acceptable ranges for each respective quality parameter.

In addition, for process constraints: The blending ratio of each individual coal type, x_j , must lie within specific bounds, denoted as $x_{j\min}$ and $x_{j\max}$, depending on industrial and operational limitations. These constraints ensure that certain coal types are used within their specified range in the blend, maintaining the desired properties of both the blend and the produced coke.

2.3. BPNN-based prediction of coke quality

Four independent BPNNs are built to predict the laboratory indices M_{40} , M_{10} , CRI and CSR of an arbitrary coal blend. The approach deliberately uses *single-output* models because preliminary tests revealed that multi-output architectures reduced accuracy for the smaller targets.

Input layer. For every quality indicator $\phi_k \in \{M_{40}, M_{10}, CRI, CSR\}$ the input vector is

$$\mathbf{x}_k = [\overline{V_{daf}}, \overline{G}, \overline{Y}, \overline{\phi_k}]^T, \quad \tilde{\psi} = \sum_{i=1}^n x_i \psi_i, \quad (8)$$

where x_i denotes the mass fraction of coal i in the blend, ψ_i the corresponding property of this coal, and $\tilde{\phi_k}$ the weighted average of the laboratory value $\phi_{k,i}$ measured on the coke produced from single coal i . All four features are normalised to zero mean and unit variance; the scaling parameters are stored and reused during inference.

Network architecture. A coarse Latin-hypercube search over layer sizes $h \in \{6, 8, 10, 12\}$, activation functions $\sigma \in \{\tanh, \text{ReLU}\}$ and L_2 regularisation factors $\lambda \in \{10^{-5}, 10^{-4}, 10^{-3}\}$ selected the compact 4–8–1 topology shown in Eq. (9) with $\sigma = \tanh$ and $\lambda = 10^{-4}$ for all four models:

$$\mathbf{h}_k = \sigma(W_k^{(1)} \mathbf{x}_k + \mathbf{b}_k^{(1)}), \quad (9a)$$

$$\hat{\phi}_k = W_k^{(2)} \mathbf{h}_k + \mathbf{b}_k^{(2)}. \quad (9b)$$

Glorot normal initialisation is applied to all weight matrices.

Training protocol. Each network is trained on $N = 332$ laboratory 40 kg oven tests using a 70 %/15 %/15 % split (train/validation/test). The mean-squared error

$$\mathcal{L}_k = \frac{1}{N} \sum_{n=1}^N (\hat{\phi}_{k,n} - \phi_{k,n})^2 \quad (10)$$

is minimised by Adam ($\eta = 1 \times 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, batch size 64). Early stopping with a patience of 10 epochs prevents over-fitting; training typically converges in 80–110 epochs.

Predictive performance. On the unseen test data the four models achieve coefficients of determination $R_{\text{test}}^2 \geq 0.90$ and root-mean-square errors below 2.0 % of the respective laboratory means.

Deployment. During coal-blend optimisation each candidate $\mathbf{x}$ is forwarded through the four networks to obtain

$$\left[\hat{M}_{40}(\mathbf{x}), \hat{M}_{10}(\mathbf{x}), \widehat{\text{CRI}}(\mathbf{x}), \widehat{\text{CSR}}(\mathbf{x}) \right]^T,$$

thereby replacing the empirical surrogates of Eqs. (4)–(7) with a fully data-driven, non-linear estimator that captures complex coal-quality interactions while remaining computationally inexpensive.

2.4. Coal blending optimization model

In coal blending optimization, it is essential to quantitatively evaluate the contribution of each single coal to the properties of the final coke product. Drawing inspiration from the NBA player evaluation system, which normalizes diverse performance metrics to allow fair comparison among athletes, we adopt a normalization strategy for all objective functions in the optimization model. This approach addresses the reality that each coal, when included in a blend, may contribute differently to critical quality parameters, and that the indicators themselves (e.g., M_{40} , M_{10} , CRI, CSR) may have different units and ranges.

The multi-objective optimization problem is formulated as:

$$\min_{\mathbf{x}} \tilde{\mathbf{F}}(\mathbf{x}) = (\tilde{f}_1(\mathbf{x}), \tilde{f}_2(\mathbf{x}), \tilde{f}_3(\mathbf{x}), -\tilde{f}_4(\mathbf{x}), \tilde{f}_5(\mathbf{x}), \tilde{f}_6(\mathbf{x}), -\tilde{f}_7(\mathbf{x}))^T,$$

where the normalized objective functions are:

$$\tilde{f}_i(\mathbf{x}) = \frac{y_{i,\text{blend}} - y_{i,\text{target}}}{y_{i,\text{target}}},$$

with $y_{i,\text{blend}}$ and $y_{i,\text{target}}$ denoting the predicted and target values of the i -th property, respectively. For maximization objectives such as M_{40} and CSR, a negative sign is used so that all objectives are handled in minimization form, ensuring consistency in the solution process.

This normalization not only puts different metrics on the same scale, but also makes explicit the impact of each coal on each target property within the optimization results. By expressing all objectives as relative deviations from their targets, it becomes possible to visualize and compare the trade-offs between cost and the full spectrum of coke quality requirements, while also highlighting the effect of each single coal's inclusion in the blend.

The selection of target values $y_{i,\text{target}}$ follows industry technical standards or user-specified operational requirements. These targets serve as reference points that define the desired performance boundaries. Through sensitivity analysis, we find that reasonable variations in target values do affect the optimal blending results, often leading to different coal selection or proportioning schemes. This underlines the role of targets as adjustable levers to align blending optimization with changing plant priorities or market demands.

The optimization process is subject to the following constraints:

$$\sum_{j=1}^n x_j = 100 \%, \quad x_j \geq 0,$$

$$y_{i,\text{blend}} \in [y_{i,\text{min}}, y_{i,\text{max}}], \quad x_j \in [x_{j,\text{min}}, x_{j,\text{max}}] \quad \forall j,$$

where x_j is the mass fraction of the j -th coal, and $y_{i,\text{blend}}$ must fall within the allowable range for each performance index.

By incorporating normalized objective functions into the coal blending optimization model, we are able to more fairly and transparently represent the trade-offs among cost, ash, sulfur, mechanical strength, and reactivity, while allowing for flexible adaptation to specific production or market contexts. This methodology ensures that the model not only reflects real-world operational complexity, but also facilitates interpretable and actionable optimization outcomes.

Algorithm 1: Pseudo-code of NSGA-III Algorithm.

Input: Population size N , Maximum iterations T_{max} , Crossover probability p_c , Mutation probability p_m , Number of reference points N_r

Output: Final population P_{final}

- 1 Initialize parameters: population size N , maximum iterations T_{max} , mutation probability p_m , crossover probability p_c , and reference points N_r ;
- 2 Generate initial population P_0 ;
- 3 Generate N_r reference points using a uniform distribution;
- 4 Assign reference points to individuals in P_0 based on their objective values;
- 5 **for** $t = 1$ **to** T_{max} **do**
- 6 Evaluate objective function values for individuals in P_t ;
- 7 Perform non-dominated sorting on P_t ;
- 8 Compute crowding distances for individuals in each front;
- 9 Select parents P_{parent} using a tournament method;
- 10 Apply crossover and mutation to P_{parent} with probabilities p_c and p_m , creating temporary population P_{temp} ;
- 11 Assign reference points to individuals in P_{temp} based on their objective values;
- 12 From P_{temp} , select new population P_{t+1} based on reference point assignment and crowding distance;
- 13 Evaluate objective function values for individuals in P_{t+1} ;
- 14 Perform non-dominated sorting based on these values;
- 15 Fill the next generation population P_{t+1} with individuals based on rank, crowding distance, and reference point assignment;
- 16 **end**
- 17 **return** the final population P_{final} ;

3. The proposed QNSGAIII algorithm

3.1. Standard NSGA-III

NSGA-III extends NSGA-II for many-objective optimization by introducing uniformly distributed reference points in the objective space [55, 56]. These reference points guide selection to maintain diversity and achieve uniform coverage of the Pareto front, addressing the challenge of balancing convergence and diversity in high-dimensional objective problems. Each generation involves non-dominated sorting, crowding distance computation, and assignment of individuals to reference points, with population updates based on rank, crowding, and proximity to these points. This approach ensures a well-distributed set of Pareto-optimal solutions.

The pseudo-code of the NSGA-III algorithm is presented in Algorithm 1, which outlines the process from initialization to the refinement of the final population through iterative generations. The ability of NSGA-III to handle many-objective problems efficiently makes it a powerful tool for complex optimization tasks where traditional multi-objective optimization methods might struggle.

3.2. Q-learning enhanced NSGA-III

Although NSGA-III is effective for many-objective optimization, its fixed P_c and P_m limits adaptability during the search. These static settings are applied throughout all generations, regardless of changes in population diversity or convergence status. As a result, the algorithm may struggle to maintain a balance between exploration and exploitation, which is particularly problematic when optimizing dynamic or high-dimensional problems [57,59].

To overcome these limitations, we propose a QNSGAIII, where a Q-learning agent is embedded in the evolutionary process to adaptively control the variation operators' parameters. The main idea is to allow

Algorithm 2: Q-learning enhanced NSGA-III (QNSGAIII).

Input: Population size N , reference points N_r , generations G , State set S , Action set $\mathcal{A}$ (discrete (P_c, P_m) pairs), Learning rate δ , discount factor η , exploration rate ϵ

Output: Final population P_G

- 1 Initialise Q-table $Q(s, a)$ for all $s \in S, a \in \mathcal{A}$;
- 2 Generate N_r reference points uniformly in the objective space.
- 3 Randomly generate initial population P_0 of size N ;
- 4 **for** $t = 0$ to $G - 1$ **do**
- 5 Extract state s_t from P_t (e.g., diversity, non-dominated front, generation index);
- 6 Select action a_t (specific (P_c, P_m)) via ϵ -greedy policy on $Q(s_t, a)$;
- 7 Apply P_c, P_m in crossover and mutation to generate offspring P_{off} ;
- 8 Merge P_t and P_{off} to form a combined population;
- 9 Evaluate objective values, perform non-dominated sorting, and assign reference points;
- 10 Select next generation P_{t+1} of size N using NSGA-III's procedure;
- 11 Compute reward r_t based on improvements in convergence and diversity;
- 12 Extract next state s_{t+1} from P_{t+1} ;
- 13 Update Q-table:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \delta [r_t + \eta \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t)];$$
- 14 **end**
- 15 **return** Final population P_G .

the algorithm to select the most suitable combination of P_c and P_m at each generation according to the current state of the population, thereby promoting a more flexible and responsive search strategy.

In our implementation, the Q-learning agent represents the state s by extracting features from the population, such as population diversity (measured by the spread of objective values or crowding distance), the number of non-dominated individuals, and the generation index. The action set $\mathcal{A}$ is composed of discrete pairs of crossover and mutation probabilities, e.g., $(P_c, P_m) \in \{(0.9, 0.1), (0.8, 0.2), \dots, (0.6, 0.4)\}$, corresponding to different exploration and exploitation intensities.

At the beginning of each generation, the agent observes the current state s_t and selects an action a_t —that is, a particular pair of P_c and P_m —according to an ϵ -greedy policy based on the current Q-table. This approach ensures that the algorithm maintains a balance between exploiting well-performing settings and exploring alternative parameter combinations.

Once the action is chosen, the algorithm applies the selected (P_c, P_m) in generating offspring via crossover and mutation operators. After offspring are produced, environmental selection is performed using the standard NSGA-III process: the current and offspring populations are combined, evaluated, and sorted; individuals are assigned to reference points; and the best solutions are selected for the next generation.

A key aspect of QNSGAIII is the design of the reward signal used for Q-learning updates. In our approach, the reward r_t is defined as a weighted combination of the change in the number of non-dominated individuals, improvement in diversity (e.g., spread or crowding), and reduction in IGD/HV metrics, reflecting both convergence and diversity gains between two generations. The Q-table is updated at each generation using the following rule:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \delta \left[r_t + \eta \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \right], \quad (11)$$

where δ is the learning rate and η the discount factor. This process enables the agent to reinforce settings that yield better search progress and adapt to the evolving landscape.

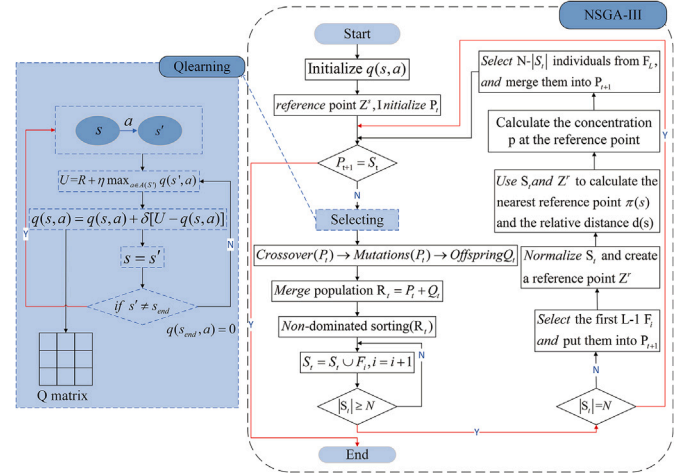


Fig. 1. Flowchart of the Q-learning enhanced NSGA-III algorithm.

Compared with the static-parameter NSGA-III, QNSGAIII enables the evolutionary search to respond to the needs of each generation. When the agent detects low diversity or stagnation, it can increase P_m to encourage exploration; as the population approaches the Pareto front, it may reduce P_m to focus on exploitation, thus preserving high-quality solutions.

The complete process is shown in Algorithm 2. Each step corresponds to an explicit operation in our MATLAB implementation, ensuring reproducibility and transparency.

Fig. 1 illustrates the overall process of QNSGAIII, which equips NSGA-III with a robust self-adaptive mechanism that automatically adjusts crossover and mutation probabilities based on real-time population feedback. This design enhances the search process by promoting diversity when needed and accelerating convergence as the algorithm progresses, yielding improved optimization performance across various many-objective scenarios.

4. Evaluation of the proposed QNSGAIII

4.1. Test function settings and performance evaluation

The DTLZ and WFG test functions [63] are commonly used to evaluate multi-objective optimization algorithms due to their complexity and varying difficulty levels. These functions, widely adopted in benchmark problems, provide a robust means of testing the performance of multi-objective optimization algorithms across different problem characteristics. The DTLZ family of problems is designed to test algorithms' ability to handle complex Pareto fronts and high-dimensional objective spaces, while the WFG functions are known for their ability to simulate problems with a wide range of difficulty levels, enabling the evaluation of the algorithms' robustness and adaptability.

To assess the performance of the proposed QNSGAIII algorithm, we use these test functions with a range of parameter settings. The key parameters, including the number of objectives (M), decision variables (D), population size, and crossover/mutation probabilities, are summarized in Table 1. These parameters influence the difficulty of the optimization task and are crucial for evaluating the algorithm's convergence and diversity. Specifically, the QNSGAIII algorithm is compared to five state-of-the-art multi-objective algorithms: MOEA/D, PESA-II, RVEA, NSGA-II, and NSGA-III. The parameters include P_c , P_m , and other relevant settings for each algorithm, as shown in Table 2. By doing so, we aim to showcase the advantages of the Q-learning enhancement in addressing the challenges posed by these test problems.

To evaluate the effectiveness of the algorithm, we employ two standard metrics: IGD and HV.

Table 1

Parameter settings and performance evaluation criteria for DTLZ and WFG test functions.

Test function	M	D	Population size	Max generations	Crossover probability	Mutation probability	Reference point (HV)	Ideal point (IGD)
DTLZ1	3	7	100	10,000	0.9	1/ D	(1, 1, 1)	(0, 0, 0)
DTLZ2	3	12	100	10,000	0.9	1/ D	(1, 1, 1)	(0, 0, 0)
DTLZ3	3	12	100	10,000	0.9	1/ D	(1, 1, 1)	(0, 0, 0)
DTLZ4	3	12	100	10,000	0.9	1/ D	(1, 1, 1)	(0, 0, 0)
DTLZ5	3	12	100	10,000	0.9	1/ D	(1, 1, 1)	(0, 0, 0)
DTLZ6	3	12	100	10,000	0.9	1/ D	(1, 1, 1)	(0, 0, 0)
DTLZ7	3	22	100	10,000	0.9	1/ D	(1, 1, 1)	(0, 0, 0)
WFG1	3	12	100	10,000	0.9	1/ D	(2, 4, 6)	(0, 0, 0)
WFG2	3	12	100	10,000	0.9	1/ D	(2, 4, 6)	(0, 0, 0)
WFG3	3	12	100	10,000	0.9	1/ D	(2, 4, 6)	(0, 0, 0)
WFG4	3	12	100	10,000	0.9	1/ D	(2, 4, 6)	(0, 0, 0)
WFG5	3	12	100	10,000	0.9	1/ D	(2, 4, 6)	(0, 0, 0)
WFG6	3	12	100	10,000	0.9	1/ D	(2, 4, 6)	(0, 0, 0)
WFG7	3	12	100	10,000	0.9	1/ D	(2, 4, 6)	(0, 0, 0)
WFG8	3	12	100	10,000	0.9	1/ D	(2, 4, 6)	(0, 0, 0)
WFG9	3	12	100	10,000	0.9	1/ D	(2, 4, 6)	(0, 0, 0)

Table 2

Parameter settings for six multi-objective optimization algorithms.

Algorithm	P_c	P_m	Crossover strategy	Mutation strategy	Special parameters
MOEA/D	0.9	0.1	Simulated Binary Crossover (SBX)	Polynomial Mutation	Neighborhood Size = 20, Weighting = 0.9
PESA-II	0.8	0.1	Simulated Binary Crossover (SBX)	Polynomial Mutation	Fitness-based Selection = True, Dominated Sorting = True
RVEA	0.8	0.1	Simulated Binary Crossover (SBX)	Polynomial Mutation	Reference Points = 100, Maximum Diversity = True
NSGA-II	0.8	0.1	Simulated Binary Crossover (SBX)	Polynomial Mutation	Tournament Size = 2, Crowding Distance = True
NSGA-III	0.8	0.1	Simulated Binary Crossover (SBX)	Polynomial Mutation	Reference Points = 100, Crowding Distance = True
QNSGA-III	0.8	0.1	Simulated Binary Crossover (SBX)	Polynomial Mutation	Learning Rate = 0.1, Discount Factor = 0.9

Table 3

IGD performance comparison of QNSGAIII against MOEA/D, PESAI, RVEA, NSGAII, and QNSGAIII on DTLZ and WFG test problems.

Problem	M	D	MOEA/D	PESAI	RVEA	NSGAII	NSGAIII	QNSGAIII
DTLZ1	3	7	3.7606e-2 (5.73e-2) =	5.3541e-2 (1.35e-1) =	5.3239e-1 (4.70e-1) -	3.1986e-1 (3.01e-1) =	1.7268e-1 (2.91e-1) =	7.3099e-2 (2.87e-1)
DTLZ2	3	12	5.4484e-2 (1.07e-5) +	7.1774e-2 (9.41e-3) -	5.4586e-2 (2.42e-4) -	7.1772e-2 (2.32e-3) -	5.4510e-2 (2.09e-5) =	5.4498e-2 (1.51e-5)
DTLZ3	3	12	1.1364e+0 (1.94e+0) -	6.0796e-2 (2.86e-2) +	3.6673e-2 (1.01e-3) +	6.8953e-2 (2.37e-3) +	1.0301e+0 (1.37e+0) -	3.6476e-2 (1.13e-4)
DTLZ4	3	12	5.4150e-1 (8.91e-1) =	7.0884e-2 (6.64e-3) -	5.5296e-2 (6.77e-4) =	7.2505e-2 (5.38e-3) -	5.5022e-2 (3.33e-4) =	5.5013e-2 (4.87e-1)
DTLZ5	3	12	3.2559e-2 (3.01e-4) -	1.4572e-2 (2.36e-3) -	8.0554e-2 (8.96e-3) -	6.7529e-3 (2.30e-4) +	1.1248e-2 (2.09e-3) =	1.2184e-2 (2.24e-3)
DTLZ6	3	12	3.1406e-2 (4.60e-3) -	1.4379e-2 (1.11e-3) +	9.7728e-2 (3.49e-2) -	6.7710e-3 (4.59e-4) +	1.8553e-2 (8.48e-3) =	1.7691e-2 (3.67e-3)
DTLZ7	3	22	1.5506e-1 (1.15e-2) -	1.0431e-1 (3.09e-1) =	2.1602e-1 (3.58e-2) -	9.1850e-2 (7.36e-3) =	1.0079e-1 (1.50e-2) -	8.9586e-2 (8.42e-3)
WFG1	3	12	6.9472e-1 (1.68e-1) =	5.5520e-1 (9.06e-2) +	7.5092e-1 (2.07e-1) =	4.9232e-1 (1.08e-1) +	7.4331e-1 (1.41e-1) =	6.6510e-1 (1.37e-1)
WFG2	3	12	3.1666e-1 (4.41e-2) -	2.0870e-1 (1.55e-2) -	2.1060e-1 (2.80e-2) -	2.3863e-1 (2.92e-2) -	1.7041e-1 (8.28e-3) =	1.6988e-1 (6.83e-3)
WFG3	3	12	2.4926e-1 (7.06e-2) -	2.8476e-1 (3.50e-1) -	2.5406e-1 (2.95e-2) -	1.3573e-1 (1.66e-2) +	1.6115e-1 (1.09e-2) =	1.6043e-1 (3.57e-2)
WFG4	3	12	2.9367e-1 (2.73e-2) -	3.0519e-1 (1.72e-2) -	2.6438e-1 (1.06e-2) -	2.9311e-1 (2.75e-2) -	2.3122e-1 (2.95e-3) =	2.3028e-1 (2.78e-3)
WFG5	3	12	2.6705e-1 (1.53e-2) -	3.0863e-1 (2.31e-2) -	2.6159e-1 (6.09e-3) -	2.8942e-1 (5.70e-3) -	2.3828e-1 (1.28e-3) -	2.3527e-1 (1.57e-3)
WFG6	3	12	3.4977e-1 (1.68e-2) -	3.6030e-1 (3.89e-2) -	3.2368e-1 (1.46e-2) -	3.3995e-1 (2.81e-2) -	2.7622e-1 (2.01e-2) =	2.7361e-1 (9.75e-3)
WFG7	3	12	3.7952e-1 (3.33e-2) -	3.0740e-1 (2.23e-2) -	2.7457e-1 (2.10e-2) -	2.9596e-1 (2.29e-2) -	2.2974e-1 (1.75e-3) =	2.3021e-1 (4.37e-3)
WFG8	3	12	3.8555e-1 (1.94e-2) -	4.1695e-1 (3.62e-2) -	3.7304e-1 (3.37e-2) -	3.8546e-1 (1.27e-2) -	3.2212e-1 (4.41e-3) =	3.1627e-1 (1.45e-2)
WFG9	3	12	3.6919e-1 (7.94e-2) -	2.9998e-1 (8.28e-3) -	2.6163e-1 (1.42e-2) -	2.8750e-1 (1.56e-2) -	2.3664e-1 (7.01e-3) -	2.3252e-1 (3.90e-3)
$\pm$ /=			1/12/3	3/11/2	1/13/2	5/9/2	0/4/12	

IGD: The IGD measures how close the solutions obtained by the algorithm are to the true Pareto front. It quantifies the average Euclidean distance between each solution in the approximate Pareto front and the closest point in the true Pareto front. A smaller IGD value indicates a better approximation, suggesting that the algorithm is capable of producing solutions with lower errors relative to the true Pareto front.

HV: The HV metric assesses the overall distribution of the solutions by computing the volume dominated by the Pareto front. It is calculated by considering the reference point and the volume enclosed by the obtained Pareto front. A higher HV value implies a more evenly spread set of solutions, reflecting the algorithm's capacity to explore the objective space more effectively and achieve a better convergence to the true Pareto front.

In the following sections, we will analyse the performance of QNSGAIII using these metrics and compare the results obtained with those of other state-of-the-art multi-objective optimisation algorithms.

4.2. Experimental results

The experiments were conducted on the PlatEMO platform, with each algorithm run for 30 independent trials to ensure statistical reliability [63,64]. Tables 3 and 4 show the mean IGD and HV values with variances, providing insights into convergence and diversity across different test problems. The following is a comprehensive analysis of QNSGAIII's performance on these test problems.

4.3. Performance on DTLZ and WFG problems

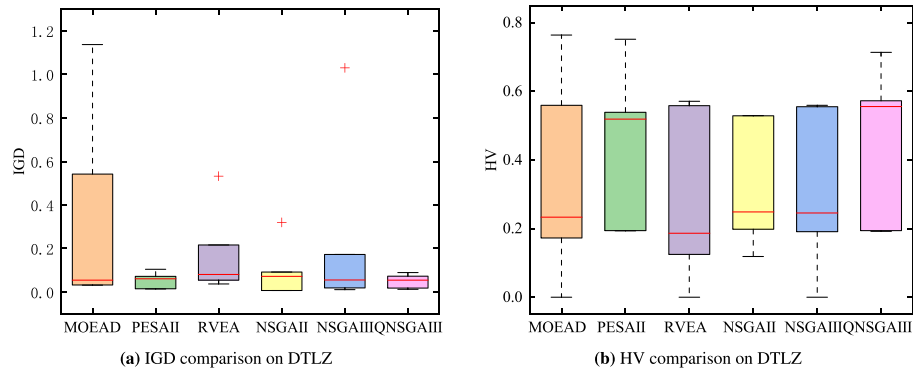
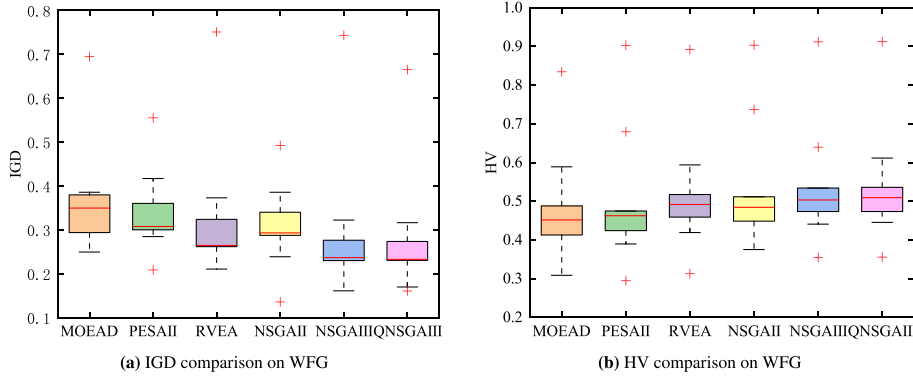
QNSGAIII is evaluated using standard benchmark functions (DTLZ and WFG series) to assess its convergence and diversity performance against other established algorithms (MOEA/D, PESAI, RVEA, NSGAII, and NSGAIII).

On the DTLZ series, QNSGAIII demonstrates robust performance, particularly excelling in DTLZ2, DTLZ3, and DTLZ4. It achieves superior IGD and HV metrics, indicating an effective balance between

Table 4

HV performance comparison of QNSGAIII against MOEAD, PESAI, RVEA, NSGAII, and QNSGAIII on DTLZ and WFG test problems.

Problem	M	D	MOEAD	PESAI	RVEA	NSGAII	NSGAIII	QNSGAIII
DTLZ1	3.0	7.0	7.6412e-1 (1.86e-1) =	7.5147e-1 (3.81e-1) =	0.0000e+0 (2.33e-2) -	1.1841e-1 (7.11e-1) =	3.8132e-1 (6.95e-1) =	7.1328e-1 (6.76e-1)
DTLZ2	3.0	12.0	5.5899e-1 (1.58e-4) =	5.1873e-1 (1.96e-2) -	5.5838e-1 (4.80e-4) -	5.2900e-1 (4.29e-3) -	5.5899e-1 (1.12e-4) -	5.5911e-1 (1.66e-4)
DTLZ3	3.0	12.0	0.0000e+0 (0.00e+0) =	5.3901e-1 (3.59e-2) +	5.7064e-1 (4.00e-3) +	5.2868e-1 (7.39e-3) +	0.0000e+0 (3.55e-2) -	5.7263e-1 (1.81e-3)
DTLZ4	3.0	12.0	3.4195e-1 (4.65e-1) =	5.3039e-1 (1.19e-2) -	5.5285e-1 (1.38e-3) -	5.2879e-1 (6.25e-3) -	5.5505e-1 (9.65e-4) =	5.5553e-1 (2.17e-1)
DTLZ5	3.0	12.0	1.8264e-1 (4.48e-4) -	1.9325e-1 (1.35e-2) =	1.4674e-1 (1.57e-2) -	1.9810e-1 (5.43e-4) +	1.9352e-1 (1.16e-3) =	1.9372e-1 (2.39e-3)
DTLZ6	3.0	12.0	1.7238e-1 (1.35e-2) -	1.9420e-1 (2.50e-3) +	1.2452e-1 (3.03e-2) -	1.9902e-1 (2.30e-4) +	1.9109e-1 (3.05e-3) =	1.9187e-1 (1.60e-3)
DTLZ7	3.0	22.0	2.3310e-1 (1.93e-2) -	2.4485e-1 (2.90e-2) =	1.8643e-1 (2.60e-2) -	2.4869e-1 (9.17e-3) =	2.4535e-1 (9.48e-3) -	2.5264e-1 (6.74e-3)
WFG1	3.0	12.0	5.8910e-1 (1.00e-1) =	6.7961e-1 (5.46e-2) +	5.9398e-1 (8.88e-2) =	7.3718e-1 (4.39e-2) +	6.3965e-1 (8.30e-2) =	6.1137e-1 (8.33e-2)
WFG2	3.0	12.0	8.3411e-1 (2.13e-2) -	9.0277e-1 (8.53e-3) -	8.9185e-1 (1.25e-2) -	9.0295e-1 (5.17e-3) -	9.1178e-1 (1.02e-2) =	9.1212e-1 (3.82e-3)
WFG3	3.0	12.0	3.0854e-1 (2.99e-2) -	2.9466e-1 (9.65e-2) -	3.1283e-1 (1.35e-2) -	3.7531e-1 (6.56e-3) +	3.5466e-1 (7.63e-3) =	3.5566e-1 (2.17e-2)
WFG4	3.0	12.0	4.8788e-1 (1.88e-2) -	4.7464e-1 (1.49e-2) -	5.1647e-1 (4.07e-3) -	5.0536e-1 (1.13e-2) -	5.2810e-1 (2.46e-3) -	5.3157e-1 (7.36e-3)
WFG5	3.0	12.0	4.8163e-1 (4.55e-3) -	4.5447e-1 (9.57e-3) -	4.9171e-1 (1.13e-3) -	4.8075e-1 (4.72e-3) -	5.0290e-1 (4.41e-3) -	5.0815e-1 (7.03e-3)
WFG6	3.0	12.0	4.3628e-1 (1.68e-2) -	4.2394e-1 (1.31e-2) -	4.5880e-1 (8.01e-3) =	4.4870e-1 (2.45e-2) -	4.7335e-1 (1.01e-2) =	4.7368e-1 (1.13e-2)
WFG7	3.0	12.0	4.5192e-1 (2.42e-2) -	4.6790e-1 (1.36e-2) -	5.1731e-1 (1.05e-2) -	5.1143e-1 (8.66e-3) -	5.3434e-1 (1.63e-3) =	5.3602e-1 (5.85e-3)
WFG8	3.0	12.0	4.0351e-1 (1.67e-2) -	3.8948e-1 (1.08e-2) -	4.1924e-1 (1.59e-2) -	4.2475e-1 (3.87e-3) -	4.4069e-1 (4.94e-3) -	4.4559e-1 (3.98e-3)
WFG9	3.0	12.0	4.1290e-1 (8.51e-2) -	4.6238e-1 (1.21e-2) -	4.9127e-1 (6.89e-3) =	4.8443e-1 (1.79e-2) -	5.0332e-1 (8.78e-3) -	5.0935e-1 (6.99e-3)
$\pm$ /=			0/11/5	3/10/3	1/12/3	5/9/2	0/7/9	

**Fig. 2.** Comparison of six algorithms—MOEAD, PESAI, RVEA, NSGAII, NSGAIII, and QNSGAIII—on DTLZ test functions.**Fig. 3.** Comparison of six algorithms—MOEAD, PESAI, RVEA, NSGAII, NSGAIII, and QNSGAIII—on WFG test functions.

convergence and diversity. Notably, QNSGAIII attains the lowest IGD value in DTLZ3 (3.6476×10^{-2}) and competitive HV values, reflecting its adeptness at handling problems with higher dimensional complexity. Nevertheless, QNSGAIII faces challenges in maintaining optimal performance in DTLZ5 and DTLZ6, suggesting that further refinement in managing highly nonlinear Pareto fronts is necessary.

For the WFG series, QNSGAIII also maintains strong performance, excelling particularly in WFG1, WFG2, and WFG3, where it delivers both leading IGD (e.g., 1.6988×10^{-1} in WFG2) and HV (e.g., 9.1212×10^{-1} in WFG2) metrics. Its performance remains solid but relatively less dominant in complex instances such as WFG4 to WFG9, where algorithms like MOEAD occasionally outperform in IGD.

4.4. Discussion on DTLZ and WFG test functions

The performance of QNSGAIII across both the DTLZ and WFG test functions, as shown in Figs. 2 and 3, illustrates its strong ability to balance convergence and diversity. In the DTLZ series, QNSGAIII consistently leads in HV, highlighting its effective exploration of the objective space and its ability to maintain a diverse solution set. This is particularly evident in the comparison of HV values in Fig. 2(b), where QNSGAIII outperforms other algorithms, demonstrating its capacity to cover a larger portion of the Pareto front. Furthermore, the relatively stable IGD results (Fig. 2(a)) indicate that QNSGAIII achieves consistent convergence across multiple runs, with minimal variation in solution quality.

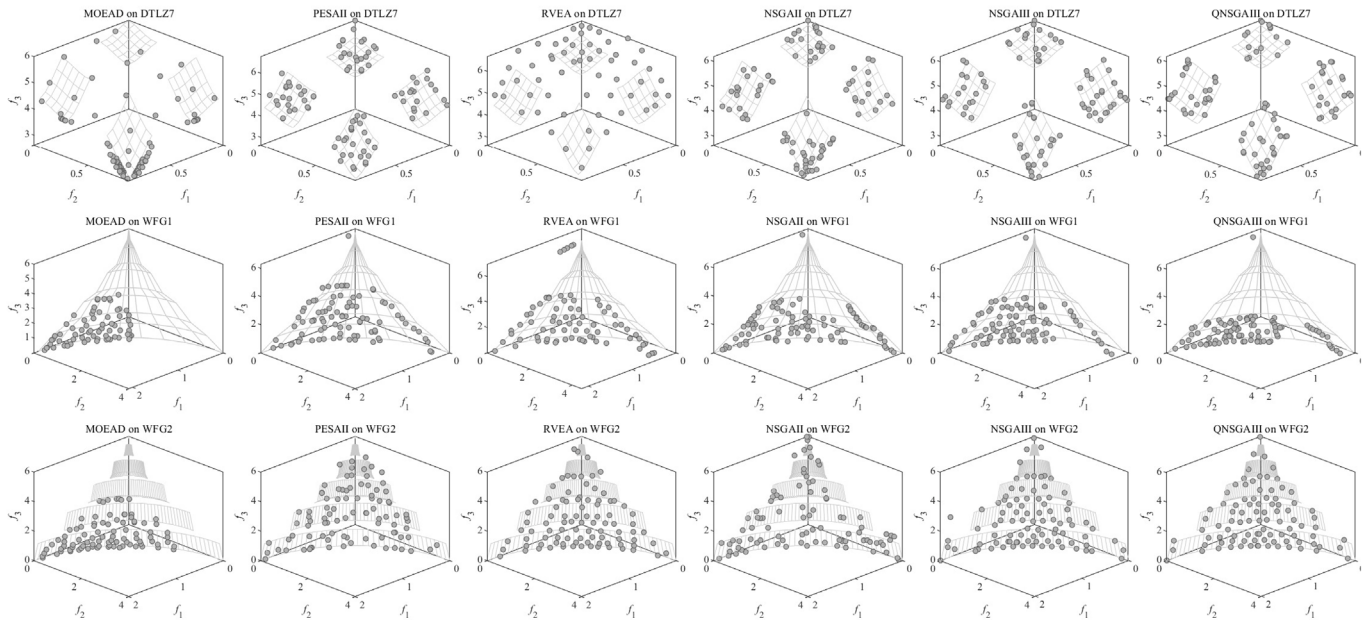


Fig. 4. Population (objective space) comparison of six algorithms—MOEAD, PESAI, RVEA, NSGAII, NSGAIII, and QNSGAIII—on selected DTLZ and WFG test functions (DTLZ7, WFG1, and WFG2).

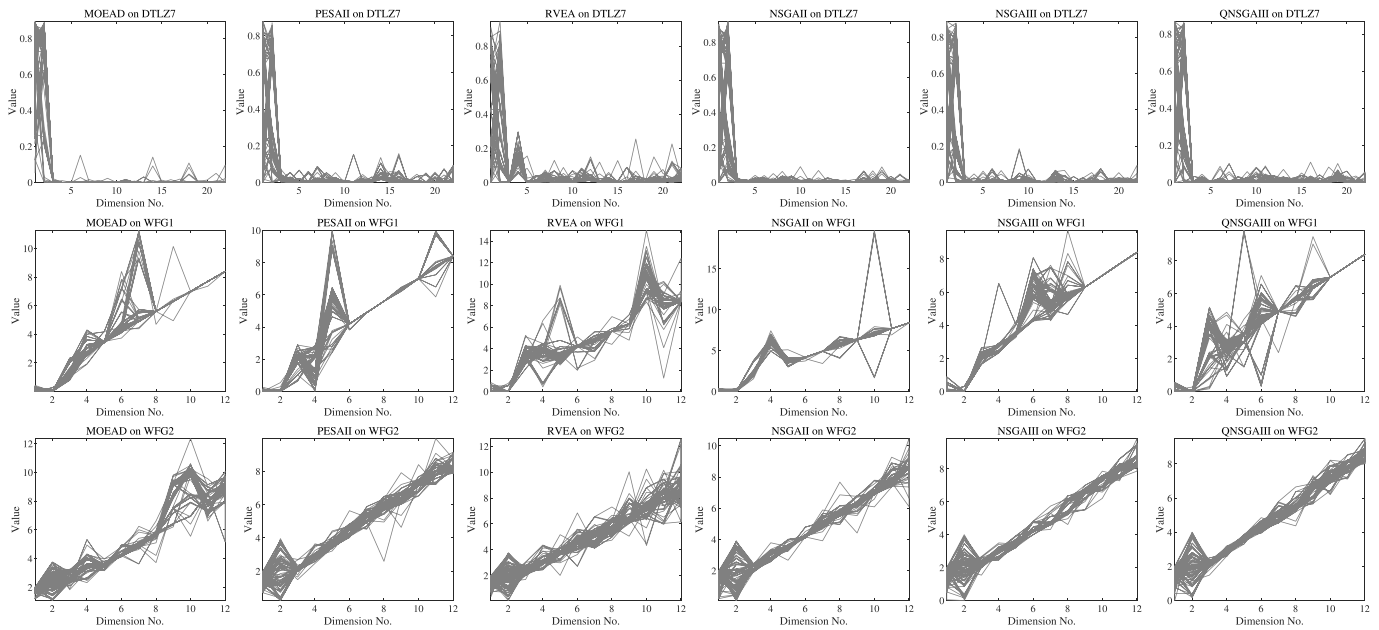


Fig. 5. Population (decision space) comparison of six algorithms—MOEAD, PESAI, RVEA, NSGAII, NSGAIII, and QNSGAIII—on selected DTLZ and WFG test functions (DTLZ7, WFG1, WFG2).

In contrast, on the WFG test functions (Fig. 3), while QNSGAIII maintains a strong HV performance, its IGD values are slightly less competitive in comparison to PESAI, which exhibits better convergence in some cases. However, QNSGAIII still performs admirably, particularly in maintaining a high diversity of solutions while achieving competitive convergence, as shown in Figs. 3(a) and 3(b). This demonstrates the algorithm's well-rounded performance across both convergence and diversity, though a more refined approach might be needed for specific cases like WFG2 where the algorithm struggles with high-dimensional problems.

Figs. 4 and 5 offer further insights into QNSGAIII's performance in the objective and decision spaces. In the objective space (Fig. 4),

QNSGAIII exhibits a more even distribution of solutions along the Pareto front compared to other algorithms, such as MOEAD and PESAI, which tend to cluster in certain regions. This highlights QNSGAIII's ability to better explore the entire Pareto front. Similarly, in the decision space (Fig. 5), the population distribution remains evenly spread, reflecting its robust performance in both convergence and diversity across decision variables.

It is important to note that, compared to the other five benchmark algorithms, QNSGAIII inevitably incurs higher computational costs and longer runtimes. This additional cost arises primarily from the integration of the Q-learning agent within the evolutionary search process. In each generation, the algorithm must perform extra steps such as

Table 5
Metallurgical coke quality classification standards based on GB/T 1996–2017.

Property Grade	A_d (%)	$S_{t,d}$ (%)	M_{40} (%)	M_{10} (%)	CRI (%)	CSR (%)
Grade 1	≤ 12.0	≤ 0.70	≥ 82.0	≤ 7.0	≤ 30	≥ 60
Grade 2	≤ 13.5	≤ 0.90	≥ 78.0	≤ 8.5	≤ 35	≥ 55
Grade 3	≤ 15.0	≤ 1.10	≥ 74.0	≤ 10.5	≤ 40	≥ 50

state extraction, action selection, Q-table updates, and reward computation. While these operations are relatively lightweight compared to evaluating the full population, they accumulate over thousands of generations, especially in large population sizes and high-dimensional objective spaces typical of DTLZ and WFG test functions.

Moreover, QNSGAIII introduces additional computational overhead through its dynamic adjustment of crossover and mutation probabilities based on real-time population feedback. This adaptive mechanism requires frequent calculations of diversity indicators—such as crowding distance and non-dominated counts—as well as convergence metrics like IGD and HV improvements. Compared to static-parameter variants such as NSGA-III, these steps add notable complexity. In our MATLAB implementation, for example, completing a typical run on a 3-objective, 12-variable WFG problem requires approximately 1.5 to 2 times longer than the baseline NSGA-III under identical population and generation settings. This relative increase becomes even more pronounced in industrial-scale problems with more objectives or tighter constraints.

While QNSGAIII shows excellent performance in many of the DTLZ and WFG test problems, there are still areas for improvement. In particular, its performance on certain high-dimensional problems, such as WFG2, reveals some inconsistencies in simultaneously maintaining both convergence and diversity. This suggests that QNSGAIII could benefit from enhanced diversity-preserving strategies, especially when tackling more complex multi-objective problems with higher-dimensional objective spaces.

5. Application of the QNSGAIII on a coking plant case study

In an industrial coking plant, coal blending optimization aims to reduce production costs while guaranteeing compliance with strict coke quality standards. This process requires the final coal blend to meet technical constraints for three grades of metallurgical coke, as outlined in Table 5. The optimal blending scheme is determined by considering the properties and market prices of available coal resources, followed by experimental validation in pilot and production-scale coke ovens. Blending technicians then select the scheme that achieves the lowest cost while fulfilling all quality requirements, as specified in GB/T 1996–2017.

The optimization process focuses on minimizing A_d , $S_{t,d}$, and CRI, while maximizing CSR and ensuring that M_{40} and M_{10} satisfy grade-specific standards. These indicators directly impact coke performance in steelmaking and the operational efficiency of the coking process.

In practice, the accuracy of coke quality prediction is critical for implementing the optimal blending scheme. A case study from a major coking plant sets strict requirements for the allowable deviation between predicted and measured values from both 40 kg laboratory ovens and production-scale coke ovens. The accepted average absolute errors are as follows:

- For coal blend indicators:
 - A_d : ≤ 0.2 %
 - $S_{t,d}$: ≤ 0.05 % (for $S_{t,d}$ between 1.5 % and 4 %, ≤ 0.1 %)
 - V_{daf} : ≤ 0.5 % (if $V_{daf} > 40$ %, ≤ 0.8 %)
 - G : ≤ 3
 - Y : ≤ 2 mm
- For coke quality indicators:
 - M_{40} : ≤ 1.5 %
 - M_{10} : ≤ 0.5 %

- CRI: ≤ 1.5 %
- CSR: ≤ 2.0 %
- A_d : ≤ 0.2 %
- $S_{t,d}$: ≤ 0.05 %

These plant-specific standards ensure that the predicted results from the coal blending model are sufficiently reliable for industrial application. By consistently achieving errors within these tolerances, the QNSGAIII-based optimization framework can be robustly applied to guide real-world blending decisions, supporting stable production of metallurgical coke that meets technical and economic targets.

5.1. Comprehensive analysis of coking properties and coal quality for 20 coal samples

A detailed evaluation was conducted on the coking properties and coal quality of 20 coal samples, classified into coking coal (JM), fat coal (FM), 1/3 coking coal (1/3JM), gas coal (QM), lean coal (SM), and non-coking coal. The analysis follows the “gas-fat-coke-lean” blending principle, which aims to optimize coke quality while maintaining cost-effectiveness in the blending process.

JM primarily provides structural integrity to coke, with sample C6 exhibiting the most favorable attributes, characterized by high coke strength ($M_{40} = 88.90$ %) and optimal reactivity indices (CRI = 15.90 %, CSR = 73.50 %). The reduced ash ($A_d = 9.83$ %) and sulfur ($S_{t,d} = 0.86$ %) contents further underline its blending suitability. FM significantly enhances coke density and binding quality; C10 shows superior caking properties ($G = 94$), excellent coke strength ($M_{40} = 81.80$ %, $M_{10} = 7.00$ %), and favorable reactivity (CRI = 16.70 %, CSR = 72.00 %). The intermediate properties of 1/3JM, exemplified by C14, yield balanced performance ($M_{40} = 83.00$ %, CRI = 23.10 %, CSR = 54.70 %, $G = 96$). QM, such as C17, controls volatile matter but has limited coke strength ($M_{40} = 69.60$ %) and suboptimal reactivity (CRI = 50.00 %, CSR = 19.00 %). SM, represented by C18, adjusts volatile content yet provides minimal coke strength ($M_{40} = 64.80$ %) and moderate reactivity (CRI = 31.50 %). Non-coking coals (e.g., C19 and C20) serve specialized roles, with C19 reducing blend sulfur ($S_{t,d} = 0.29$ %) and C20, despite its coke fines classification, enhancing mechanical coke strength marginally. Each coal type's proportion in the blend is carefully controlled, reflecting their individual contributions and limitations. The key findings from this evaluation, along with the detailed coal quality and coke analysis for each sample, are presented in Table 6.

5.2. Coke quality prediction

As established in Section 2, the A_d and $S_{t,d}$ of a coal blend are governed by near-linear mixing laws and are thus calculated as weighted averages of the constituent coals. In contrast, hot-strength and reactivity indices—including M_{40} , M_{10} , CRI, and CSR—are influenced by a broader range of interacting variables and exhibit significant non-linearity, making simple linear prediction inadequate. To address these complexities, four back-propagation neural networks (BPNNs, architecture 3–8–1) were trained, each dedicated to one specific target index.

Prior research has shown that V_{daf} , G , and Y are the most influential coal properties for predicting coke quality. These parameters form the basis of the input layer for each BPNN. To further incorporate the indirect effects of mineral content and other minor components, the input set was expanded to include the blend-weighted values of M_{40} , M_{10} , CRI, and CSR for each single coal. This strategy integrates the effects of ash chemistry and related properties into the nonlinear predictor, without causing excessive growth of the parameter space.

The performance of the coke quality prediction model was evaluated using data from various blending schemes, tested both in 40 kg laboratory ovens and in production-scale ovens. Table 7 summarizes the measured, predicted, and absolute error values for all coke quality and coal blend indicators across different grades. In 40 kg tests, the prediction errors for M_{40} , M_{10} , CRI, CSR, A_d , and $S_{t,d}$ consistently met

Table 6
Coal quality and coke analysis for 20 coal samples (tested in 40 kg top-charged coke oven).

Coal Name Index	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20
Coal Price (RMB/ton)	526.00	520.00	520.00	456.00	578.00	577.00	541.00	496.00	515.00	849.00	375.00	763.50	495.00	440.00	535.00	480.00	431.00	515.00	420.00	120.00
Coal A_d (%)	9.72	10.20	10.00	9.05	10.56	9.83	9.52	8.81	9.76	9.82	9.62	9.21	9.90	9.56	6.87	8.16	10.69	9.22	4.02	11.46
Coal $S_{t,d}$ (%)	1.12	1.34	1.27	1.80	0.69	0.86	0.90	1.20	0.92	0.53	2.30	0.55	0.72	1.31	0.42	0.45	0.70	0.39	0.29	0.88
Coal V_{daf} (%)	22.12	18.95	25.86	22.39	17.85	19.52	32.58	33.58	33.06	32.82	35.76	36.62	35.55	35.82	35.06	30.30	36.15	17.09	38.30	8.82
Coal G	81.00	78.00	78.00	85.00	81.00	80.00	90.00	87.00	90.00	94.00	91.00	92.70	87.00	96.00	75.00	82.00	68.00	66.00	NA	NA
Coal X (mm)	36.50	30.00	45.00	45.00	25.00	32.50	36.00	53.00	25.00	26.80	47.50	43.20	46.50	28.70	52.50	31.50	53.00	28.00	NA	NA
Coal Y (mm)	17.50	12.00	16.00	22.00	12.50	15.50	23.00	25.50	24.00	26.00	28.00	23.60	21.50	23.00	13.00	25.00	13.00	10.00	NA	NA
Coke M_{40} (%)	87.70	82.70	85.80	86.30	87.20	88.90	81.00	85.40	82.50	81.80	78.60	74.80	75.20	83.00	67.80	78.60	69.60	64.80	NA	NA
Coke M_{10} (%)	6.50	12.40	7.60	8.00	8.30	7.40	10.40	8.80	10.30	7.00	12.30	13.30	11.80	7.80	14.60	10.60	13.60	26.40	NA	NA
Coke CRI (%)	18.30	25.40	35.70	21.50	18.60	15.90	24.90	25.50	32.40	16.70	33.60	28.50	25.40	23.10	56.40	25.30	50.00	31.40	NA	NA
Coke CSR (%)	70.20	51.50	37.20	60.00	65.50	73.50	60.10	58.80	48.00	72.00	35.80	45.30	53.80	54.70	20.30	65.00	19.00	38.80	NA	NA
Coke A_d (%)	12.48	12.23	12.90	11.29	12.85	12.04	12.73	12.41	13.29	13.90	13.75	13.86	13.56	10.04	11.40	15.38	13.81	11.46	5.23	NA
Coke $S_{t,d}$ (%)	0.99	1.18	1.16	1.63	0.58	0.75	0.70	0.95	0.78	0.45	1.85	0.44	0.67	1.17	0.42	0.36	0.56	0.41	0.24	NA
Classification	JM	JM	JM	JM	JM	JM	FM	FM	FM	FM	FM	1/3JM	1/3JM	1/3JM	1/3JM	1/3JM	QM	SM	RN	Non-Coking Coal

industrial tolerance requirements. For grades 1 to 5, the deviations were generally below 1.5 % for M_{40} , 0.5 % for M_{10} , 1.5 % for CRI, and 2 % for CSR, confirming the high predictive accuracy of the BPNN model. Errors for other blend indicators such as V_{daf} , G , and Y also remained minor, with variations largely attributable to intrinsic differences in coal properties and test conditions.

As shown in Table 8, the mean absolute error (MAE) for M_{40} , M_{10} , CRI, CSR, A_d , and $S_{t,d}$ demonstrates consistently high accuracy, with low standard deviations (STD) reflecting reliable performance across different blending schemes. While slightly larger errors are observed in production-scale oven data, the deviation remains within acceptable ranges, confirming the transferability of the BPNN approach from laboratory to industrial settings. These results indicate that the BPNN predictor provides a reliable foundation for multi-objective coal blending optimization and supports robust decision-making for coke production.

The validated accuracy of the coke quality prediction model provides a solid basis for subsequent optimization. In the next section, the proposed QNSGAI algorithm is employed to derive optimal blending strategies that meet both technical and economic criteria.

5.3. Verification of cost optimization for coal blending

In practical industrial production, coal blending is tailored to produce three grades of metallurgical coke, with each grade subject to specific quality requirements as summarized in Table 5. In this context, A_d and $S_{t,d}$ are critical indicators, and their maximum allowable values are set according to industrial standards to ensure the final coke quality. For operational reliability, we follow the common practice of not specifying an absolute lower bound for these indicators; instead, the minimum is set at 95 % of the upper limit. For example, if the maximum permitted ash content is 12.0 %, the minimum constraint is set at 11.4 %. Similarly, if the upper limit for total sulfur is 0.70 %, the lower bound is defined as 0.665 %. This strategy prioritizes achieving the required coke quality while encouraging further reductions in impurity levels, within reasonable operational margins.

The full list of quality constraints for the optimization problem, as applied in this study, is detailed below. For each coke grade, the quality indicators are constrained as follows:

- Ash content (A_d): $A_{d,min} = 0.95 \times A_{d,max}$, $A_{d,max}$ as specified in Table 5.
- Sulfur content ($S_{t,d}$): $S_{t,d,min} = 0.95 \times S_{t,d,max}$, $S_{t,d,max}$ per the table.
- Mechanical strength and reactivity indices (M_{40} , M_{10} , CRI, CSR): respective upper and lower bounds strictly follow the grade requirements listed in Table 5.
- The sum of the blending ratios for all coal types must equal 100 %, with the proportion of each individual coal (x_j) constrained within operational limits: $x_{j,min} \leq x_j \leq x_{j,max}$.

All constraint parameters are reported explicitly in Table 5 for full transparency and reproducibility.

Following the above criteria, the optimized blending ratios for each coke grade are presented in Table 9, and the corresponding quality results are shown in Table 10. The results demonstrate that each optimized blend satisfies the upper and lower bounds of all key quality indicators, confirming the effectiveness of the multi-objective optimization model in balancing cost control and coke quality. These findings provide a robust reference for practical coal blending operations in the coking industry.

5.3.1. Grade 1 metallurgical coke

For Grade 1 coke, the optimized blend predominantly utilizes high-quality coking coals (C5, C6, C16) with a moderate addition of fat coal (C7), supplemented by smaller amounts of 1/3 coking coal (C12, C13, C15) and lean coal (C18). The detailed proportions and roles of each component are as follows:

Table 7

Summary of measured, predicted, and absolute error values for all coke quality and blend indicators.

Sample / Indicator	Coke quality indicators						Blend indicators					
	M_{40} (%)	M_{10} (%)	CRI (%)	CSR (%)	A_d (%)	$S_{t,d}$ (%)	V_{daf} (%)	G	X	Y	A_d (%)	$S_{t,d}$ (%)
tolerance	≤1.50	≤0.50	≤1.50	≤2.00	≤0.20	≤0.05	≤0.50	≤3.00	–	≤2.00	≤0.20	≤0.05
grade1-40 kg-test	85.80	8.40	29.40	54.90	12.10	0.69	25.93	78.00	39.50	17.00	9.28	0.82
grade1-40 kg-predict	84.86	8.76	27.19	54.35	12.44	0.70	25.65	77.47	37.88	16.55	9.51	0.81
grade1-40 kg-error	0.94	0.36	2.21	0.55	0.34	0.01	0.28	0.53	1.62	0.45	0.23	0.01
grade1-oven-test	84.50	7.70	26.10	66.90	12.38	0.67	25.31	80.00	25.30	17.80	9.60	0.79
grade1-oven-predict	84.54	7.53	25.72	67.14	12.35	0.67	24.62	77.63	37.85	15.61	9.50	0.76
grade1-oven-error	0.04	0.17	0.38	0.24	0.03	0.00	0.69	2.37	12.55	2.19	0.10	0.03
grade2-40 kg-test	84.60	10.40	36.20	40.10	12.39	0.78	27.50	77.00	35.00	16.50	8.98	0.84
grade2-40 kg-predict	84.00	10.53	35.96	40.19	12.52	0.72	27.93	72.11	41.97	16.98	9.32	0.84
grade2-40 kg-error	0.60	0.13	0.24	0.09	0.13	0.06	0.43	4.89	6.97	0.48	0.34	0.00
grade2-oven-test	83.30	8.10	28.90	62.20	12.97	0.71	28.03	73.00	23.70	14.70	9.89	0.79
grade2-oven-predict	81.61	8.13	29.05	61.59	12.91	0.69	27.07	71.35	40.21	14.45	9.72	0.78
grade2-oven-error	1.69	0.03	0.15	0.61	0.06	0.02	0.96	1.65	16.51	0.25	0.17	0.01
CSR-A-40 kg-test	86.50	7.90	23.60	61.80	11.99	0.66	23.88	73.00	31.00	19.00	9.39	0.77
CSR-A-40 kg-predict	86.78	7.93	23.76	61.04	11.99	0.66	24.75	79.45	34.18	18.08	9.37	0.77
CSR-A-40 kg-error	0.28	0.03	0.16	0.76	0.00	0.00	0.87	6.45	3.18	0.92	0.02	0.00
CSR-B-40 kg-test	77.30	13.20	27.00	54.90	12.91	0.72	25.50	74.00	36.00	15.50	9.48	0.79
CSR-B-40 kg-predict	78.81	13.31	26.82	54.25	12.33	0.70	26.52	76.80	38.03	16.83	9.42	0.81
CSR-B-40 kg-error	1.51	0.11	0.18	0.65	0.58	0.02	1.02	2.80	2.03	1.33	0.06	0.02
CSR-C-40 kg-test	77.60	14.70	35.10	37.80	13.05	0.78	28.76	71.00	42.00	14.00	9.56	0.84
CSR-C-40 kg-predict	76.50	14.70	34.96	38.01	12.53	0.72	29.64	72.03	42.22	16.80	9.28	0.85
CSR-C-40 kg-error	1.10	0.00	0.14	0.21	0.52	0.06	0.88	1.03	0.22	2.80	0.28	0.01

Table 8

Statistical errors for coke quality and blend indicators.

Metric	M_{40}	M_{10}	CRI	CSR	A_d	$S_{t,d}$	V_{daf}	G	Y
MAE	0.8800	0.1186	0.4943	0.4443	0.2371	0.0136	0.7036	2.5314	1.0736
MSE	1.0957	0.0270	0.7409	0.2549	0.1063	0.0003	0.7068	9.1737	2.1273
STD	0.6122	0.1231	0.7611	0.2591	0.2416	0.0234	0.5721	2.3062	1.0553

- **C5 (15 %):** This coal features low ash ($A_d = 10.56$ %) and low sulfur ($S_{t,d} = 0.69$ %), contributing directly to structural stability in the final coke.
- **C6 (16 %):** Serving as the primary blend component, C6 provides outstanding strength ($M_{40} = 88.90$ %) and favorable reactivity (CRI = 15.90 %, CSR = 73.50 %).
- **C16 (19 %):** With robust strength ($M_{40} = 78.60$ %) and reactivity characteristics (CRI = 25.30 %, CSR = 65.00 %), C16 supports the coke's mechanical skeleton.
- **C7 (9 %):** This fat coal, notable for a high caking index ($G = 90$), supplies critical binding properties essential for coke formation.
- **C15 (10 %):** The 1/3 coking coal in the blend helps regulate volatile matter ($V_{daf} = 35.06$ %), but its dosage is carefully controlled to preserve overall strength.

The resulting coke satisfies the Grade 1 requirements specified in GB/T 1996–2017:

- $M_{40} = 82.07$ %, meeting and slightly exceeding the minimum threshold.
- $M_{10} = 7.00$ %, well within the standard limit for wear resistance.
- CRI = 24.83 %, safely below the maximum allowed value (≤ 30 %).
- CSR = 60.42 %, above the required 60 %, indicating solid post-reaction durability.

The production cost for Grade 1 coke is 540.90 RMB/ton. While this is higher than for lower grades, it is offset by improved mechanical and reactive performance, making it especially suitable for high-efficiency blast furnace operations. Moreover, optimizing the blend—by reducing the share of premium coking and fat coals by 3 %—can lower costs by approximately 10 RMB/ton, achieving a balance between economic and technical performance without sacrificing coke quality.

5.3.2. Grade 2 metallurgical coke

For Grade 2 coke, the blend is designed for cost-efficiency by increasing the share of standard coking coals (C1, C4, C5) and fat coals (C8, C9), complemented by smaller proportions of 1/3 coking coals (C14, C15, C16). Key blend components and their functions are summarized as follows:

- **C1 (13 %):** Offers reliable strength ($M_{40} = 87.70$ %) and moderate reactivity (CRI = 18.30 %, CSR = 70.20 %).
- **C4 (7 %):** Characterized by low ash ($A_d = 9.05$ %) and sulfur ($S_{t,d} = 1.80$ %), supporting structural consistency.
- **C8 (13 %):** A fat coal with moderate caking index ($G = 87$), improving coke cohesion.
- **C14 (10 %):** Functions as a 1/3 coking coal, contributing to binding and stability.
- **C16 (13 %):** Another 1/3 coking coal, managing volatile content ($V_{daf} = 30.30$ %).

The resulting coke meets all Grade 2 specifications:

- $M_{40} = 82.14$ %, above the standard minimum of 78.0 %.
- $M_{10} = 8.49$ %, marginally higher than the 8.5 % threshold but still within industrial tolerance.
- CRI = 31.01 %, well below the 35 % upper limit.
- CSR = 55.07 %, providing sufficient post-reaction strength.

Grade 2 coke is produced at 499.87 RMB/ton, striking a favorable balance between price and technical performance, and is well-suited to medium-scale blast furnace use. Compared to higher grades, optimizing the blend by reducing the share of high-quality coking and fat coals by 4 % can decrease the cost by approximately 15 RMB/ton, offering further economic benefits without compromising key properties.

Table 9
Optimized coal blending ratios for metallurgical coke grades.

Coal Name Scheme	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20
Grade 1(CoalBlend1)	14 %	0 %	0 %	7 %	15 %	16 %	9 %	0 %	0 %	3 %	0 %	1 %	1 %	0 %	10 %	19 %	0 %	5 %	0 %	0 %
Grade 2(CoalBlend2)	13 %	0 %	8 %	7 %	1 %	8 %	4 %	13 %	11 %	0 %	1 %	0 %	0 %	10 %	7 %	13 %	3 %	0 %	0 %	1 %
Grade 3(CoalBlend3)	10 %	0 %	1 %	14 %	0 %	0 %	1 %	7 %	0 %	0 %	15 %	0 %	0 %	15 %	0 %	19 %	17 %	0 %	0 %	1 %

5.3.3. Grade 3 metallurgical coke

Grade 3 coke adopts a low-cost strategy, utilizing a combination of standard coking coals (C1, C4), fat coal (C10), 1/3 coking coals (C14, C16), and gas coal (C17). The blend is formulated as follows:

- **C1 (10 %):** A cost-effective coking coal, providing essential structural support.
- **C4 (14 %):** Features low ash ($A_d = 9.05 \%$) and sulfur ($S_{t,d} = 1.04 \%$), enhancing the coke matrix.
- **C10 (15 %):** A fat coal with a high caking index ($G = 94$), improving coke densification.
- **C14 (15 %):** This 1/3 coking coal enhances the blend's strength and stability.
- **C17 (17 %):** Gas coal is used to regulate volatile matter ($V_{daf} = 35.76 \%$), contributing to proper plasticity.

The final product satisfies Grade 3 technical criteria:

- $M_{40} = 79.61 \%$, exceeding the required minimum of 74.0 %.
- $M_{10} = 10.48 \%$, within the limit of 10.5 %.
- $CRI = 33.56 \%$, under the 40 % threshold.
- $CSR = 49.98 \%$, meeting the minimum specification for Grade 3 coke.

The cost of Grade 3 coke is 450.19 RMB/ton, making it the most economical option among the three grades. This blend is suitable for scenarios where cost control takes precedence over maximum strength and reactivity. Reducing the share of high-quality coking and fat coals by 5 % further lowers costs by around 20 RMB/ton, ensuring economic advantage without sacrificing compliance with coke quality standards.

5.4. Comparative analysis with other algorithms

The effectiveness of QNSGAIII in solving the coal blending optimization problem was evaluated against several established multi-objective algorithms, including MOEAD, PESAI, RVEA, NSGAI, and NSGAI. The comparative analysis was performed using two widely recognized performance metrics: the IGD and HV, as illustrated in Fig. 6. Detailed results are presented in Tables 11 and 12, highlighting each algorithm's ability to approximate the Pareto front and to maintain diversity among the generated solutions—both crucial for optimal coal blending across different coke grades.

IGD Analysis: QNSGAIII consistently achieves superior convergence, as reflected by the lowest IGD values across all test cases. For CoalBlend1, QNSGAIII records an IGD of 1.2858×10^0 , outperforming MOEAD and RVEA, whose IGD values are notably higher. This lower IGD indicates that QNSGAIII identifies solutions closely aligned with the true Pareto front, particularly excelling in balancing low sulfur and volatile matter content while maximizing CSR for Grade 1 coke. Such performance is essential for meeting the rigorous quality standards required in metallurgical coke production.

QNSGAIII also outperforms the other algorithms for CoalBlend2 and CoalBlend3, achieving the lowest IGD values in each case. This demonstrates that QNSGAIII maintains strong convergence toward the true Pareto front, regardless of the flexibility in quality constraints across different grades, and balances objectives such as maximizing CSR and minimizing CRI.

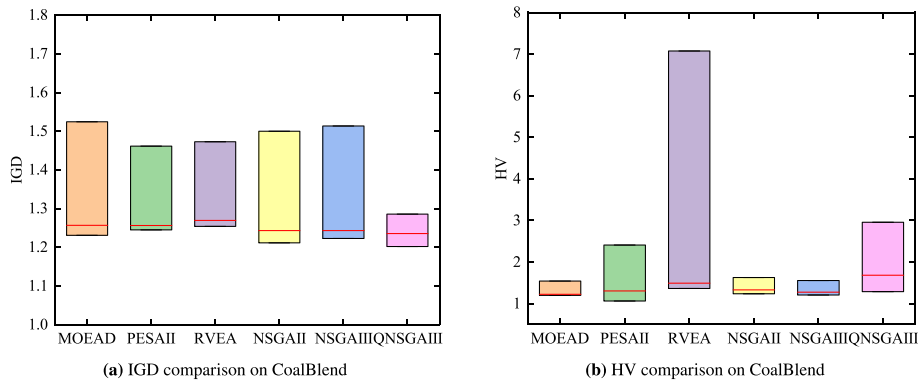
HV Analysis: The HV metric evaluates the diversity and extent of the solution set. In Table 12, QNSGAIII attains the highest HV for CoalBlend1 (2.9582×10^0), reflecting its ability to provide both convergence and a broad spread of high-quality solutions. This characteristic is essential for practical coal blending, as it ensures that diverse and feasible blend options are available for decision-making.

Although RVEA achieves a high HV for CoalBlend2, its performance on other test problems is less consistent compared to QNSGAIII, highlighting QNSGAIII's advantage in producing diverse and high-quality solutions across all scenarios. Algorithms such as MOEAD and NSGAI typically achieve lower HV values, suggesting limited diversity relative to QNSGAIII.

Table 10

Average coal blending and coke quality for three metallurgical coke grades.

Scheme	Unit price (RMB/ton)	A _d (%)	S _{t,d} (%)	M ₄₀ (%)	M ₁₀ (%)	CRI (%)	CSR (%)	Vdaf (%)	G	X (mm)	Y (mm)
Grade 1	540.90	11.93	0.68	82.07	7.00	24.83	60.42	25.33	81.41	34.95	18.22
Grade 2	499.87	12.32	0.88	82.14	8.49	31.01	55.07	29.11	82.82	37.42	19.88
Grade 3	450.19	12.88	1.04	79.61	10.48	33.56	49.98	30.94	82.19	39.75	20.74

**Fig. 6.** Comparison of six algorithms—MOEAD, PESAI, RVEA, NSGAII, NSGAIII, and QNSGAIII—on CoalBlend test functions.**Table 11**

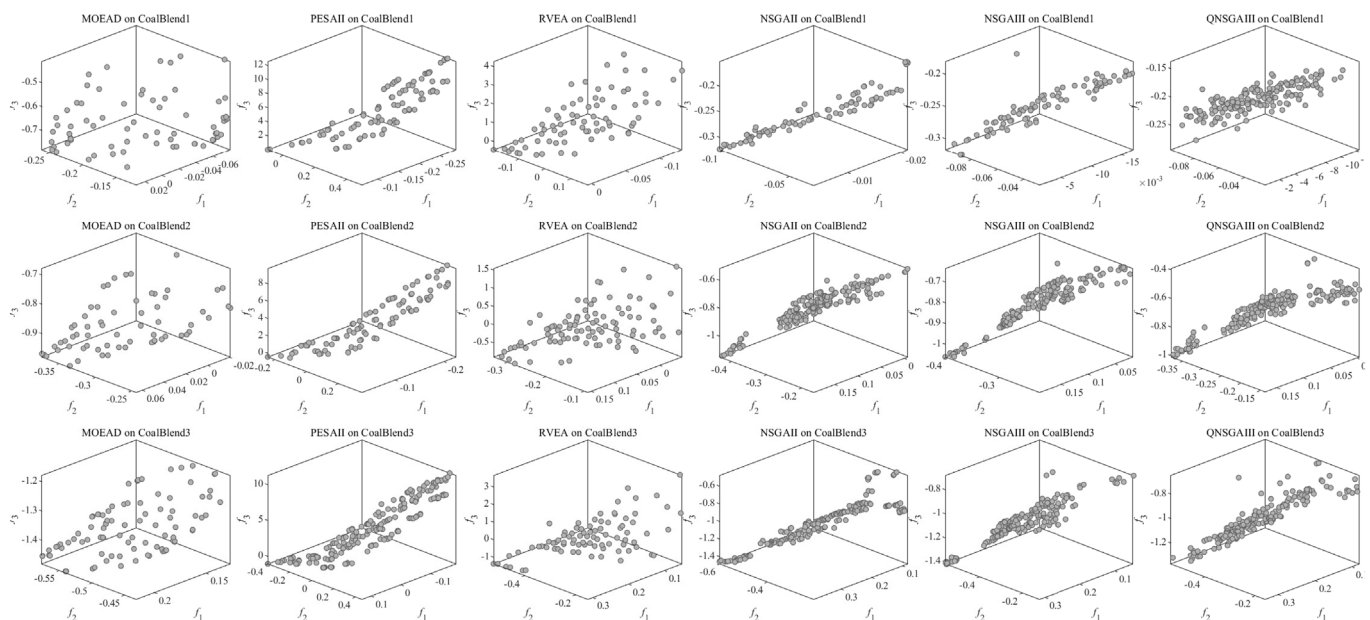
IGD values for different algorithms across three metallurgical coke grades.

Problem	M	D	MOEAD	PESAI	RVEA	NSGAII	NSGAIII	QNSGAIII
CoalBlend1	3	20	1.5245e+0 (5.76e-2) -	1.4613e+0 (2.93e-2) =	1.4725e+0 (7.28e-2) +	1.4997e+0 (5.55e-2) =	1.5134e+0 (5.30e-2) =	1.2858e+0 (1.74e-2)
CoalBlend2	3	20	1.2311e+0 (1.23e-2) -	1.2561e+0 (5.83e-3) =	1.2542e+0 (1.66e-2) -	1.2114e+0 (1.15e-2) =	1.2228e+0 (1.26e-2) =	1.2020e+0 (9.17e-3)
CoalBlend3	3	20	1.2569e+0 (6.53e-3) =	1.2451e+0 (1.54e-2) =	1.2694e+0 (7.75e-3) =	1.2432e+0 (1.28e-2) =	1.2432e+0 (4.62e-3) -	1.2355e+0 (1.46e-2)
±/=			0/2/1	0/0/3	1/1/1	0/0/3	0/1/2	

Table 12

HV values for different algorithms across three metallurgical coke grades.

Problem	M	D	MOEAD	PESAI	RVEA	NSGAII	NSGAIII	QNSGAIII
CoalBlend1	3	20	1.2256e+0 (1.38e-1) -	1.0649e+0 (7.74e-2) =	1.3639e+0 (3.47e-1) =	1.3323e+0 (1.81e-1) =	1.2778e+0 (1.31e-1) =	2.9582e+0 (7.62e-1)
CoalBlend2	3	20	1.2014e+0 (3.74e-2) =	2.4083e+0 (1.96e+0) =	7.0734e+0 (6.29e+0) =	1.2387e+0 (4.28e-2) =	1.2093e+0 (4.19e-2) =	1.2911e+0 (1.08e-1)
CoalBlend3	3	20	1.5441e+0 (6.99e-2) =	1.3056e+0 (1.24e-1) -	1.4924e+0 (9.36e-2) =	1.6290e+0 (1.11e-1) =	1.5568e+0 (7.80e-2) =	1.6862e+0 (1.31e-1)
±/=			0/1/2	0/1/2	0/0/3	0/0/3	0/0/3	

**Fig. 7.** Pareto fronts of coal blending optimization using six multi-objective algorithms.

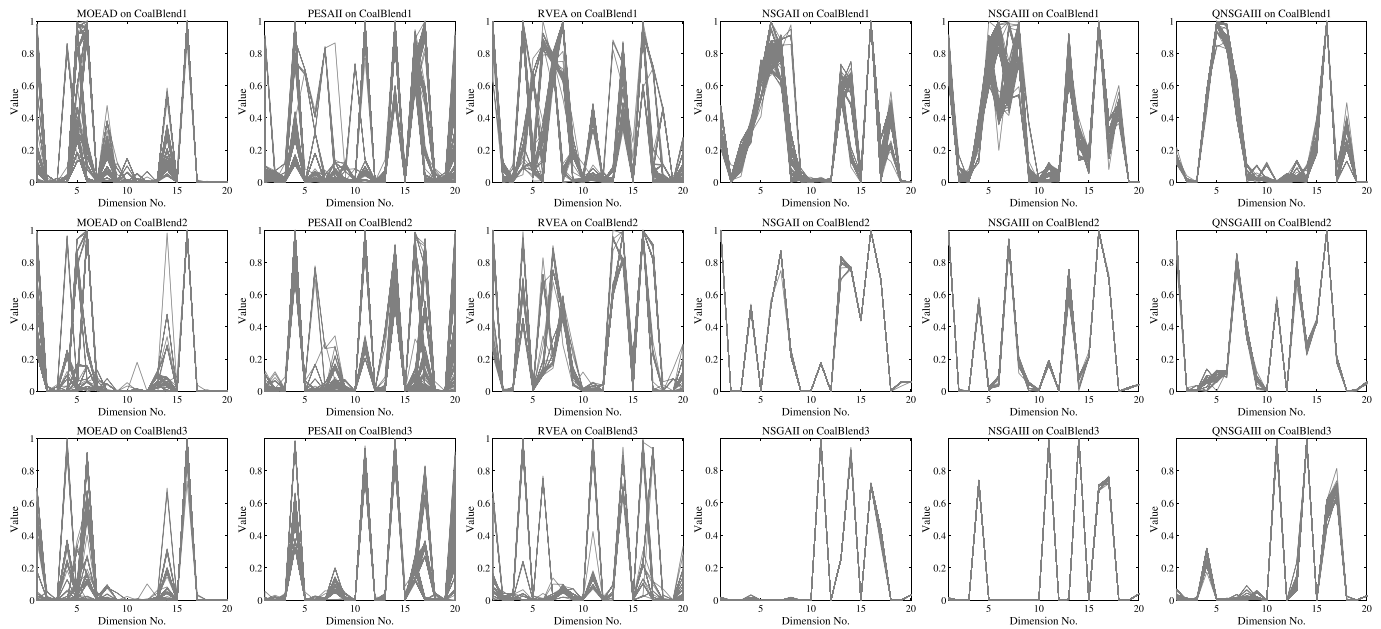


Fig. 8. Solution distribution in the decision space for three metallurgical coke grades using different optimization algorithms.

Optimization Relevance in Coal Blending: In coal blending, the objective is to achieve blends that meet specific requirements for A_d , $S_{t,d}$, CSR, and CRI, while minimizing cost. The results in Tables 11 and 12 show that QNSGAIII delivers the best overall trade-off, providing blends that meet or exceed technical specifications for each grade at a competitive cost.

Pareto Front and Solution Distribution: Fig. 7 displays the Pareto fronts obtained by each algorithm. QNSGAIII generates a dense cluster of solutions close to the true Pareto front, with good distribution across the objective space. This diversity allows for effective selection of blends for different production needs. For Grade 1 coke, QNSGAIII is particularly effective at minimizing sulfur and volatile content while maximizing CSR, addressing the stringent requirements of high-grade metallurgical coke.

For Grades 2 and 3, where quality criteria are less restrictive, QNSGAIII still offers diverse and competitive solutions, as seen in the spread of its Pareto front.

Solution Distribution in Decision Space: Fig. 8 illustrates the distribution of coal blend solutions in the decision space. QNSGAIII produces well-dispersed solutions, ensuring that a wide variety of coal types are incorporated in the blends. This flexibility is particularly valuable for Grade 3 coke, where the optimization strategy favors maximizing CSR and CRI even with higher A_d and $S_{t,d}$ values. In comparison, MOEAD often shows a more concentrated distribution, limiting the range of practical blend options.

Despite the extra runtime, the improvement in convergence and diversity, as evidenced by the experimental results, justifies the computational expense in many practical scenarios. For industrial coal blending cases, where the solution quality has direct economic impact, the moderate increase in computational time is often acceptable, especially when the optimization process can be parallelized or run offline. Future work may further optimize the Q-learning component or exploit parallel architectures to mitigate the runtime gap while preserving the adaptive benefits brought by the reinforcement learning mechanism.

6. Conclusion

This study establishes a novel multi-objective coal blending optimization model for metallurgical coke production, integrating seven critical objectives: minimization of blending cost, ash content (A_d),

sulfur content ($S_{t,d}$), abrasion strength (M_{10}), and coke reactivity index (CRI), alongside maximization of crushing strength (M_{40}) and coke strength after reaction (CSR). To address the NP-hard nature of the problem (solution space of approximately 100^{20} for 20 coal types), we propose QNSGAIII, a Q-learning enhanced NSGA-III algorithm. This hybrid method dynamically adjusts crossover/mutation probabilities and reference points through reinforcement learning, achieving a balance between exploration and exploitation. Benchmark tests on DTLZ and WFG functions confirm its superiority over MOEA/D, NSGA-III, and other algorithms, with average IGD and HV improvements of 12.7 % and 18.3 %, respectively.

The industrial validation using 20 single coals demonstrates QNSGAIII's practical efficacy. For three metallurgical coke grades, Pareto-optimal solutions reduce premium coking/fat coal usage by 3 %–5 % while lowering blending costs by 10–20 RMB/ton. Specifically, Grade 1 coke achieves $M_{40} = 82.07$ % and CSR = 60.42 % at 540.90 RMB/ton, meeting GB/T 1996–2017 standards. Grade 3 coke maintains compliance ($M_{40} = 79.61$ %, CSR = 49.98 %) at minimal cost (450.19 RMB/ton), highlighting the model's adaptability across quality tiers. The algorithm's ability to generate diverse blend structures (e.g., 14 % C1 + 16 % C6 for Grade 1 vs. 17 % C17 for Grade 3) ensures operational flexibility.

These findings validate the integration of reinforcement learning into evolutionary algorithms for complex industrial optimization. By simultaneously addressing cost-quality trade-offs and process constraints, QNSGAIII provides a systematic framework for intelligent cokemaking. Future work will focus on extending this approach to large-scale blending scenarios and incorporating real-time process adjustments.

CRedit authorship contribution statement

Zhongfeng Li: Methodology, Investigation, Data curation, Software, Writing - original draft. **Lei Liu:** Supervision, Project administration. **Zhenlong Zhao:** Investigation. **Hui Jin:** Data curation. **Yuting Zhuo:** Methodology, Resources, Formal analysis, Supervision, Project administration.

Declaration of competing interest

The authors declare the following financial interests/personal relationships that may be considered potential competing interests:

Zhongfeng Li reports financial support was provided by Yingkou Institute of Technology. Zhongfeng Li reports a relationship with Yingkou Institute of Technology that includes employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available upon request.

References

- [1] Ameri Siahuei MR, Ataei M, Sereshki F. Investigating relationships between intrinsic properties, preparation parameters of coal and coke quality: a systematic literature review. *Rudarsko geolosko naftni zbornik* 2024;39(4):55–76. <https://doi.org/10.17794/rgn.2024.4.5>
- [2] Zhang L, Wang G, Xue Q, Zuo H, She X, Wang J. Effect of preheating on coking coal and metallurgical coke properties: a review. *Fuel Process Technol* 2021;221:106942. <https://doi.org/10.1016/j.fuproc.2021.106942>
- [3] Appiah J, Tian L, Dou J, Chen Y, Chen X, Xu X, et al. Investigation into the impact of coal blending on the carbon structure of chars obtained from Chinese coals during coking. *J Anal Appl Pyrolysis* 2024;179:106504. <https://doi.org/10.1016/j.jaap.2024.106504>
- [4] Agrawal A. Technological advancements in cokemaking. *Miner Process Extract Metall Rev* 2024;1–16. <https://doi.org/10.1080/08827508.2023.2300113>
- [5] Al Hosni S, Domini M, Vahidzadeh R, Bertanza G. Potential and environmental benefits of biochar utilization for coal/coke substitution in the steel industry. *Energies* 2024;17(11):2759. <https://doi.org/10.3390/en17112759>
- [6] Zhuo Y, Hu Z, Shen Y. Cfd study of hydrogen injection through tuyeres into ironmaking blast furnaces. *Fuel* 2021;302:120804. <https://doi.org/10.1016/j.fuel.2021.120804>
- [7] Pang Z, Bu J, Yuan Y, Zheng J, Xue Q, Wang J, et al. The low-carbon production of iron and steel industry transition process in China. *Steel Res Int* 2024;95(3):2300500. <https://doi.org/10.1002/srin.202300500>
- [8] Ibitoye SE, Loha C, Mahamood RM, Jen T-C, Alam M, Sarkar I, et al. An overview of biochar production techniques and application in iron and steel industries. *Bioprocess* 2024;11(1):65. <https://doi.org/10.1186/s40643-024-00779-z>
- [9] Guo P, Lei R, Liu W, Liu J, Hou C, Sun Y, et al. Activated coke for synergistic removal of multiple pollutants from sintering flue gas: a review. *Fuel* 2025;381:133506. <https://doi.org/10.1016/j.fuel.2024.133506>
- [10] Diez MA, Alvarez R, Barriocanal C. Coal for metallurgical coke production: predictions of coke quality and future requirements for cokemaking. *Int J Coal Geol* 2002;50(1–4):389–412. [https://doi.org/10.1016/S0166-5162\(02\)00123-4](https://doi.org/10.1016/S0166-5162(02)00123-4)
- [11] Granda M, Blanco C, Alvarez P, Patrick JW, Menendez R. Chemicals from coal coking. *Chem Rev* 2014;114(3):1608–36. <https://doi.org/10.1021/cr400256y>
- [12] Zheng H, Xu R, Zhang J, Daghighaleh O, Schenk J, Li C, et al. A comprehensive review of characterization methods for metallurgical coke structures. *Materials* 2021;15(1):174. <https://doi.org/10.3390/ma15010174>
- [13] Jayasekara AS, Brooks B, Steel K, Koshy P, Hockings K, Tahmasebi A. Microalgae blending for sustainable metallurgical coke production—impacts on coking behaviour and coke quality. *Fuel* 2023;344:128130. <https://doi.org/10.1016/j.fuel.2023.128130>
- [14] Niu W, Li Y, Li Q, Wang J, Wang G, Zuo H, et al. Physical and chemical properties of metallurgical coke and its evolution in the blast furnace ironmaking process. *Fuel* 2024;366:131277. <https://doi.org/10.1016/j.fuel.2024.131277>
- [15] Brooks B, Rish SK, Lomas H, Jayasekara A, Tahmasebi A. Advances in low carbon cokemaking—influence of alternative raw materials and coal properties on coke quality. *J Anal Appl Pyrolysis* 2023;106083. <https://doi.org/10.1016/j.jaap.2023.106083>
- [16] Kundu N, Biswas P, Bhunia P, Ghosh R, Sarkar S. Evolution characteristics of metallurgical coals for coke making through thermogravimetric-mass spectroscopic measurements. *J Environ Chem Eng* 2021;9(6):106874. <https://doi.org/10.1016/j.jece.2021.106874>
- [17] Cui B, Shen Y, Guo J, Jin X, Wang M, Xie W, et al. A study of coking mechanism based on the transformation of coal structure. *Fuel* 2022;328:125360. <https://doi.org/10.1016/j.fuel.2022.125360>
- [18] Zhao J, Liu H, Zhang H, Song X, Zuo H, Wang G, et al. Metallurgical performance and structural characteristics of cokes of hypercoal prepared from the mixture of low-rank coal and biomass residue. *Fuel* 2023;332:126069. <https://doi.org/10.1016/j.fuel.2022.126069>
- [19] Zolotukhin YA, Andreichikov NS, Eremin AY, Kraskovskaya TF, Kuprygin VV. Formulating coal batch for coking: a review. *Coke Chem* 2021;64:97–104. <https://doi.org/10.3103/S1068364X21030108>
- [20] Nasifullina AI, Starkov MK, Gabdulkhakov RR, Rudko VA. Petroleum coking additive—raw material component for metallurgical coke production. Part 2. Experimental studies of obtaining a petroleum coking additive. *CIS Iron Steel Rev* 2022;24:9–16.
- [21] Farajzadeh E, Zhuo Y, Shen Y. Numerical investigation of the impact of injecting coke oven gas on raceway evolution in an ironmaking blast furnace. *Fuel* 2024;358:130345. <https://doi.org/10.1016/j.fuel.2023.130345>
- [22] Zhuo Y, Shen Y. Transient 3d CFD study of pulverised coal combustion and coke combustion in a blast furnace: effect of blast conditions. *Fuel* 2023;340:127468. <https://doi.org/10.1016/j.fuel.2023.127468>
- [23] Shih J-S, Frey HC. Coal blending optimization under uncertainty. *Eur J Oper Res* 1995;83(3):452–65. [https://doi.org/10.1016/0377-2217\(94\)00243-6](https://doi.org/10.1016/0377-2217(94)00243-6)
- [24] Shen Y, Hu Y, Wang M, Bao W, Chang L, Xie K. Speciation and thermal transformation of sulfur forms in high-sulfur coal and its utilization in coal-blending coking process: a review. *Chin J Chem Eng* 2021;35:70–82. <https://doi.org/10.1016/j.cjche.2021.04.007>
- [25] Adeleke AA, Nzerem P, Ayuba S, Odusote JK, Olosolo AI, Ikubanni PP, et al. Formed coke from coal and plastic: a review. In: 2023 2nd international conference on multidisciplinary engineering and applied science (icmeas); IEEE; 2023. p. 1–6. <https://doi.org/10.1109/ICMEAS58693.2023.10429867>
- [26] Patrick JW. The coking of coal. *Sci Prog* 1974;375–99. <https://www.jstor.org/stable/43420254>
- [27] Fomeni FD. A multi-objective optimization approach for the blending problem in the tea industry. *Int J Prod Econ* 2018;205:179–92. <https://doi.org/10.1016/j.ijpe.2018.08.036>
- [28] Amini SH, Vass C, Shahabi M, Noble A. Optimization of coal blending operations under uncertainty—robust optimization approach. *Int J Coal Prep Util* 2022;42(1):30–50. <https://doi.org/10.1080/19392699.2019.1574262>
- [29] Khasraw D, Martin C, Herbert J, Li Z. A comprehensive literature review of biomass characterisation and application for iron and steelmaking processes. *Fuel* 2024;368:131459. <https://doi.org/10.1016/j.fuel.2024.131459>
- [30] Kurniawan T, Irawan A, Alwan H, Hernanto R, Wahyudi W, Kodarif AR, et al. A kinetic model approach for predicting coke reactivity index from coal and coal blend properties. *Int J Coal Prep Util* 2022;42(5):1318–35. <https://doi.org/10.1080/19392699.2019.1710498>
- [31] Jenkins DR, Mahoney MR. A mechanistic model for the softening of coking coal and its use for predicting the dilatation of blends. *Fuel* 2015;153:585–94. <https://doi.org/10.1016/j.fuel.2015.02.116>
- [32] Li Z, Tie W, Huang J, Wang Q, Li T, Yang S, et al. Analysis of the differences in coke thermal properties testing. *Fuel* 2023;354:129359. <https://doi.org/10.1016/j.fuel.2023.129359>
- [33] Zhang Q, Wu X, Feng A, Shi M. Prediction of coke quality at baosteel. *Fuel Process Technol* 2004;86(1):1–11. [https://doi.org/10.1016/S0378-3820\(03\)00058-4](https://doi.org/10.1016/S0378-3820(03)00058-4)
- [34] Fu X, Pang Q, Yang X, Zhan W, He Z. Effect of high temperature on macroscopic properties and microstructure of metallurgical coke. *Fuel* 2024;356:129543. <https://doi.org/10.1016/j.fuel.2023.129543>
- [35] Jarernsuk S, Phruksaphanrat B. Solving tea blending problems using interactive fuzzy multi-objective linear programming. *Processes* 2022;11(1):49. <https://doi.org/10.3390/pr11010049>
- [36] Yan S, Lv C, Yao L, Hu Z, Wang F. Hybrid dynamic coal blending method to address multiple environmental objectives under a carbon emissions allocation mechanism. *Energy* 2022;254:124297. <https://doi.org/10.1016/j.energy.2022.124297>
- [37] Wang J, Wang W, Chen X, Bao J, Hao Q, Zheng H, et al. Optimizing the performance of iron coke by coal blending: insights from the metallurgical properties and structural characteristics. *J Clean Prod* 2024;450:142017. <https://doi.org/10.1016/j.jclepro.2024.142017>
- [38] Qiu Y, Hui Y, Zhao P, Dou J, Bhattacharya S, Dai B, et al. The interpretable deep learning-based coke quality prediction strategy aligned with domain knowledge for sustainable smart cokemaking applications. *Fuel* 2025;398:135533. <https://doi.org/10.1016/j.fuel.2025.135533>
- [39] Qiu Y, Hui Y, Zhao P, Cai C-H, Dai B, Dou J, et al. A novel image expression-driven modeling strategy for coke quality prediction in the smart cokemaking process. *Energy* 2024;294:130866. <https://doi.org/10.1016/j.energy.2024.130866>
- [40] Qiu Y, Hui Y, Zhao P, Wang M, Guo S, Dai B, et al. The employment of domain adaptation strategy for improving the applicability of neural network-based coke quality prediction for smart cokemaking process. *Fuel* 2024;372:132162. <https://doi.org/10.1016/j.fuel.2024.132162>
- [41] Yuan Y, Qu Q, Chen L, Wu M. Modeling and optimization of coal blending and coking costs using coal petrography. *Information Sciences* 2020;522:49–68. <https://doi.org/10.1016/j.ins.2020.02.072>
- [42] Liu M, Yu Z, Li B, Wang Q, Ren H, Xu D. Coal allocation optimization based on a hybrid residual prediction model with an improved genetic algorithm. *Eng Appl Artif Intell* 2024;137:109072. <https://doi.org/10.1016/j.engappai.2024.109072>
- [43] Li X, Yu R, Liu G, Chen L, Chen E, Liu S. Research on multi-objective optimization algorithm for coal blending. In: China National Conference on Big Data and Social Computing; Springer; 2023. p. 37–60. <https://doi.org/10.1007/978-981-99-3925>
- [44] Li H, Wang S, Yang Y, Fang H, Guo W. Construction of intelligent coal blending system based on bp neural network and genetic algorithm. *Ironmaking Steelmaking* 2025;03019233241312762.

- [45] Zi J, Jin F, Zhao J, Wang W, Liu Q. A multi-objective simulated annealing algorithm-based coal blending optimization approach in coking process. In: 2020 IEEE Intl Conf on Dependable, Autonomic and Secure Computing, Intl Conf on Pervasive Intelligence and Computing, Intl Conf on Cloud and Big Data Computing, Intl Conf on Cyber Science and Technology Congress (DASC/PiCom/CBDCoM/CyberSciTech); IEEE; 2020. p. 103–9. <https://ieeexplore.ieee.org/document/9251154>
- [46] Dash PS, Guha M, Chakraborty D, Banerjee PK. Prediction of coke csr from coal blend characteristics using various techniques: a comparative evaluation. Int J Coal Prep Util 2012;32(4):169–92. <https://doi.org/10.1080/19392699.2011.640301>
- [47] Tiwari HP, Banerjee PK, Saxena VK. A novel technique for assessing the coking potential of coals/coal blends for non-recovery coke making process. Fuel 2013;107:615–22. <https://doi.org/10.1016/j.fuel.2012.12.015>
- [48] Kumari P, Singh AK. Blending strategies of coals from diverse origins for efficient coke making. J Geol Soc India 2025;101(2):220–9. <https://doi.org/10.17491/jgsi/2025/174084>
- [49] Jeuken R, Forbes M, Kearney M. Optimal blending strategies for coking coal using chance constraints. J Oper Res Soc 2021;72(12):2690–703. <https://doi.org/10.1080/01605682.2020.1811167>
- [50] Liu C, Xie Z, Sun F, Chen L. Exergy analysis and optimization of coking process. Energy 2017;139:694–705. <https://doi.org/10.1016/j.energy.2017.08.006>
- [51] Zhang Q, Li H. MOEA/D: a multiobjective evolutionary algorithm based on decomposition. IEEE Trans Evol Comput 2007;11(6):712–31. <https://doi.org/10.1109/TEVC.2007.892759>
- [52] Corne DW, Knowles JD, Oates MJ. The pareto envelope-based selection algorithm for multiobjective optimization. Int Conf Parallel Problem Solving Nat 2000:839–48. <https://doi.org/10.1007/3-540-45356-382>
- [53] Chen H, Cheng R, Wen J, Li M, Yang S. A reference vector guided evolutionary algorithm for many-objective optimization. IEEE Trans Evol Comput 2015;20(5):773–91. <https://doi.org/10.1109/TEVC.2016.2519378>
- [54] Deb K, Pratap A, Agarwal S, Meyarivan TAMT. A fast and elitist multiobjective genetic algorithm: NSGA-II. IEEE Trans Evol Comput 2002;6(2):182–97. <https://doi.org/10.1109/4235.996017>
- [55] Deb K, Jain H. An evolutionary many-objective optimization algorithm using reference-point-based nondominated sorting approach, part i: solving problems with box constraints. IEEE Trans Evol Comput 2013;18(4):577–601. <https://doi.org/10.1109/TEVC.2013.2281535>
- [56] Jain H, Deb K. An evolutionary many-objective optimization algorithm using reference-point based nondominated sorting approach, part ii: handling constraints and extending to an adaptive approach. IEEE Trans Evol Comput 2013;18(4):602–22. <https://doi.org/10.1109/TEVC.2013.2281534>
- [57] Tang H, Xiao Y, Zhang W, Lei D, Wang J, Xu T. A dql-NSGA-III algorithm for solving the flexible job shop dynamic scheduling problem. Expert Syst Appl 2024;237:121723. <https://doi.org/10.1016/j.eswa.2023.121723>
- [58] Qi R, Li J-Q, Wang J, Jin H, Han Y-Y. Qmoea: a q-learning-based multiobjective evolutionary algorithm for solving time-dependent green vehicle routing problems with time windows. Inf Sci 2022;608:178–201. <https://doi.org/10.1016/j.ins.2022.06.056>
- [59] Asghari A, Sohrabi MK. Bi-objective cloud resource management for dependent tasks using q-learning and NSGA-3. J Ambient Intell Human Comput 2024;15(1):197–217. <https://doi.org/10.1007/s12652-022-03885-y>
- [60] Chen R, Wu B, Wang H, Tong H, Yan F. A q-learning based NSGA-II for dynamic flexible job shop scheduling with limited transportation resources. Swarm Evol Comput 2024;90:101658. <https://doi.org/10.1016/j.swevo.2024.101658>
- [61] Yang H, Wang Z, Gao Y, Zhou W. Bi-objective multi-mode resource-constrained multi-project scheduling using combined NSGA ii and q-learning algorithm. Appl Soft Comput 2024;152:111201. <https://doi.org/10.1016/j.asoc.2023.111201>
- [62] Li P, Xue Q, Zhang Z, Chen J, Zhou D. Multi-objective energy-efficient hybrid flow shop scheduling using q-learning and gvns driven NSGA-II. Comput Oper Res 2023;159:106360. <https://doi.org/10.1016/j.cor.2023.106360>
- [63] Tian Y, Cheng R, Zhang X, Jin Y. Platemo: a MATLAB platform for evolutionary multi-objective optimization [educational forum]. IEEE Comput Intell Mag 2017;12(4):73–87. <https://ieeexplore.ieee.org/abstract/document/8065138>
- [64] Tian Y, Zhu W, Zhang X, Jin Y. A practical tutorial on solving optimization problems via platemo. Neurocomputing 2023;518:190–205. <https://doi.org/10.1016/j.neucom.2022.10.075>